Program Komputer:

STRUKXEL

VBA Excel to ETABS Converter v1.0

Penyusun:

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1 Pendahuluan

Desain struktur gedung beton bertulang berbasis analisis respons spektrum dihitung berdasarkan parameter struktur setiap komponen dan respons dinamis struktur terhadap gempa, sehingga analisis dilakukan dengan dua tahap utama, yaitu:

a) Analisis model struktur (pemodelan geometri, material, dan pembebanan).

b) Analisis respons spektrum (analisis dinamis sesuai SNI 1726:2019).

Analisis Analisis model struktur untuk setiap komponen pekerjaan dihitung berdasarkan kebutuhan parameter yang meliputi: geometri (*grid*, jumlah lantai, tinggi lantai), dimensi elemen (kolom, balok, pelat), properti material (mutu beton, mutu baja), dan pembebanan (beban mati, beban hidup, beban gempa). Besarnya parameter untuk masing-masing elemen struktur telah banyak dikembangkan dalam berbagai standar/acuan, antara lain SNI 1726:2019, SNI 1727:2020, dan SNI 2847:2019. Selanjutnya untuk dapat melakukan analisis struktur terhadap parameter-parameter tersebut diperlukan *software* analisis seperti ETABS yang dilengkapi dengan fitur *Open Application Programming Interface* (oAPI).

Bagian kedua dari proses analisis adalah ekstraksi hasil analisis struktur dari ETABS. Hasil analisis merupakan *output* berupa periode fundamental, gaya dasar, simpangan antar lantai, gaya dalam elemen, dan rasio kapasitas elemen untuk setiap kombinasi beban. Hasil analisis tersebut dapat ditampilkan dalam berbagai format untuk evaluasi desain sesuai kebutuhan pengguna.

Berdasarkan tahapan tersebut, proses analisis memerlukan referensi parameter struktur, definisi material, pembebanan, dan respons spektrum seismik. Parameter geometri untuk setiap konfigurasi struktur relatif berbeda sehingga memerlukan *input* baru untuk konstruksi yang berbeda. Sedangkan referensi material dan pembebanan dapat digunakan kembali untuk pekerjaan desain struktur yang lain, meskipun nantinya diperlukan penyesuaian sesuai lokasi atau *update* standar. Hubungan antara parameter geometri, material, pembebanan, dan hasil analisis akan menjadi kompleks, sehingga tidak mudah jika dikerjakan secara manual.

Pekerjaan desain struktur yang kompleks terdiri atas banyak tahapan dan komponen elemen struktur, sehingga diperlukan *input* parameter yang banyak pula. Aplikasi komputer diperlukan untuk menyelesaikan pekerjaan tersebut secara cepat dan akurat. Program STRUKXEL adalah program komputer yang dikembangkan melalui integrasi VBA Excel dengan ETABS (*Extended*

Three Dimensional Analysis of Building Systems) menggunakan ETABS oAPI (*Open Application Programming Interface*). Program ini dirancang untuk mengotomatisasi proses desain struktur gedung beton bertulang tipikal berbasis analisis respons spektrum sesuai dengan SNI 1726:2019, SNI 1727:2020, dan SNI 2847:2019.

Aplikasi ini (STRUKXEL) akan menyimpan parameter material, pembebanan, dan kombinasi beban yang pernah dilakukan pada pekerjaan sebelumnya. Referensi parameter standar akan tersimpan dan dapat dijadikan acuan awal pada pekerjaan selanjutnya, sehingga tidak memerlukan definisi ulang. Sedangkan geometri struktur tentunya akan berbeda untuk setiap proyek karena bersifat *project-specific*. STRUKXEL dapat mempercepat proses analisis serta menjamin akurasi relasi antara parameter *input* yang diperlukan dalam desain struktur dan hasil analisis dari ETABS.

2 Fitur Program

Fitur program STRUKXCEL terdiri atas:

a) Laman menu utama

Laman menu utama berupa tampilan awal ketika aplikasi dibuka, dan digunakan pengguna untuk mengisi data. Laman menu utama terdiri atas 7 tombol, yaitu:

1) Tombol menu informasi proyek
2) Tombol menu data geometri struktur
3) Tombol menu dimensi struktur
4) Tombol menu material dan konfigurasi
5) Tombol menu pembebanan
6) Tombol menu parameter gempa
7) Tombol menu run.

b) Laman informasi proyek

Laman informasi proyek terdiri atas:

1) Textbox input nama proyek yang dikerjakan
2) Textbox input lokasi proyek yang dikerjakan
3) Textbox input nama pengguna yang mengerjakan

c) Laman data geometri struktur

Laman ini berfungsi untuk mendefinisikan parameter geometri struktur yang terdiri atas:

1) Textbox input jumlah grid X
2) Textbox input jumlah grid Y
3) Textbox input panjang spacing X
4) Textbox input panjang spacing Y
5) Textbox input jumlah lantai (story)
6) Textbox input tinggi lantai (story)

d) Laman dimensi struktur

Laman ini berfungsi untuk mendefinisikan parameter dimensi elemen struktur yang digunakan yang terdiri atas:

1) Textbox input lebar kolom
2) Textbox input tebal kolom
3) Textbox input lebar balok
4) Textbox input tebal balok
5) Textbox input tebal pelat lantai

e) Laman material dan konfigurasi

Laman ini berfungsi untuk mendefinisikan material yang digunakan serta konfigurasi tulangan yang digunakan. Parameter pada laman ini terdiri atas:

1) Textbox input mutu beton
2) Textbox input mutu baja
3) Textbox input diameter tulangan longitudional
4) Textbox input diameter tulangan sengkang
5) Textbox input jumlah tulangan longitudional
6) Textbox input jumlah tulangan sengkang
7) Textbox input jarak sengkang

f) Laman pembebanan

Laman ini berfungsi untuk mendefinisikan beban yang digunakan pada proyek. Parameter pada laman ini terdiri atas:

1) Textbox input beban hidup pada pelat lantai
2) Textbox input beban hidup pada pelat atap
3) Textbox input beban mati tambah keramik
4) Textbox input beban mati tambah plafon
5) Textbox input beban 1/2 pasangan bata

g) Laman parameter gempa

Laman ini berfungsi untuk mendefinisikan parameter seismik yang akan digunakan dalam analisis respons spektrum. Parameter pada laman ini terdiri atas:

1) Textbox input kelas tanah
2) Textbox input peak ground acceleration (PGA)
3) Textbox input SS MCEr
4) Textbox input S1 MCEr

5) Textbox input TL
6) Textbox input T0
7) Textbox input Ts
8) Textbox input Sds
9) Textbox input Sd1
10) Textbox input response modification factor
11) Textbox input system overstrength
12) Textbox input deflection amplification
13) Textbox input occupancy importance

h) Laman run

Laman ini berfungsi untuk menjalankan hasil dari parameter yang sudah ada sebelumnya. Laman ini terdiri atas:

1) CheckBox create model ETABS
2) CheckBox masukkan pembebanan
3) CheckBox apply response spectrum
4) CheckBox run and design check
5) CheckBox export joint reaction
6) CheckBox export base reaction
7) CheckBox export beam forces
8) CheckBox export column forces
9) CheckBox export node displacement

3 Prosedur Pengembangan

Prosedur pengembangan program computer STRUKXEL dirancang melalui pendekatan sistematis untuk memastikan terciptanya alat bantu desain yang andal, akurat, dan memenuhi standar yang berlaku. Tahapan ini merubah kebutuhan analisis secara konvensional yang kompleks menjadi algoritma komputasi yang efisien. Secara garis besar, alur pengembangan sistem ini dibagi menjadi tahapan utama, yaitu identifikasi kebutuhan, perancangan struktur sistem, implementasi algoritma, serta kalibrasi kinerja program.

3.1 Identifikasi Kebutuhan dan Studi Pustaka

Pengembangan diawali dengan analisis terhadap metode desain konvensional yang sering kali tidak efisien akibat proses iterasi manual yang berulang. Studi literatur dilakukan untuk memetakan integrasi antara Microsoft Excel sebagai antarmuka input dan ETABS sebagai mesin analisis utama melalui fitur *Open Application Programming Interface* (oAPI). Seluruh parameter desain dikalibrasi agar sesuai dengan standar nasional, meliputi SNI 1726:2019 untuk ketahanan gempa, SNI 1727:2020 untuk pembebanan, dan SNI 2847:2019 untuk material beton bertulang.

3.2 Perancangan Struktur Sistem

Sistem dirancang dengan struktur modular untuk memisahkan fungsi antarmuka pengguna, logika pemrosesan data, dan eksekusi analisis, Perancangan ini mencakup pembuatan formulir input (*UserForm*) pada Excel untuk menampung variabel geometri, material, dan beban secara terstruktur. Struktur data disusun agar kompatibel dengan objek model pada ETABS, memastikan transfer informasi berjalan tanpa error.

3.3 Implementasi Algoritma

Tahap ini merupakan inti pengembangan di mana logika desain diterjemahkan ke dalam skrip pemrograman makro. Fokus implementasi meliputi pembangunan algoritma untuk koneksi otomatis antar-aplikasi, penerjemahan data numerik dari Excel menjadi model visual 3D pada ETABS, serta otomatisasi penerapan beban statis serta dinamis. Logika pemrograman disusun untuk menangani berbagai skenario input guna meminimalkan potensi kegagalan eksekusi program.

3.4 Kalibrasi dan Pengujian Kinerja

Tahap akhir pengembangan adalah kalibrasi kinerja program menggunakan studi kasus model gedung tipikal. Kalibrasi dilakukan melalui metode komparatif, yaitu membandingkan *output* program STRUKXEL terhadap hasil perhitungan atau pemodelan manual pada ETABS. Parameter yang diuji mencakup periode getar alami, gaya geser dasar (*base shear*), simpangan antar lantai, serta gaya dalam elemen untuk menjamin akurasi dari hasil analisis.

4 Pengembangan VBA Excel

Bab ini menguraikan logika teknis dan mekanisme operasional STRUKXEL dalam mengotomatisasi proses desain struktur. Pengembangan difokuskan pada pemanfaatan ETABS oAPI untuk mengontrol fungsi analisis secara eksternal tanpa keterlibatan langsung pengguna. Uraian pengembangan pada bab ini diklasifikasikan ke dalam empat mekanisme utama, mulai dari koneksi sistem, pemodelan geometri, otomatisasi beban, hingga ekstraksi hasil analisis.

4.1 Mekanisme Koneksi dan Integrasi

Sistem dikembangkan menggunakan protokol integrasi yang mampu mendeteksi status aplikasi ETABS secara otomatis. Algoritma koneksi dirancang untuk membuka *instance* baru atau menghubungkan diri ke aplikasi yang sedang berjalan, menciptakan jembatan komunikasi data yang stabil. Mekanisme ini memungkinkan parameter yang diinput pada *UserForm* Excel dikirim dan diterjemahkan langsung oleh sistem integrasi menjadi perintah pemodelan.

4.2 Logika Pemodelan Geometri Otomatis

Logika pemodelan dibangun menggunakan algoritma pengulangan (*looping*) bertingkat untuk menyusun kerangka struktur. Sistem secara cerdas mengonversi data grid dan elevasi lantai menjadi objek kolom, balok, dan pelat pada koordinat secara presisi. Pengembangan ini mencakup definiisi otomatis properti material dan penampang elemen, serta penetapan kondisi batas seperti tumpuan jepit dan diafragma lantai kaku untuk memastikan model struktur berperilaku sesuai asumsi desain.

4.3 Otomatisasi Pembebanan dan Analisis

Modul pembebanan menerapkan logika kondisional untuk mendistribusikan beban secara spesifik. Program melakukan perbedaan terhadap beban hidup antara lantai tipikal dan atap, serta mengidentifikasi balok tepi untuk aplikasi beban dinding secara otomatis. Dalam analisis gempa, sistem mendefinisikan fungsi respons spektrum berdasarkan parameter wilayah dan membentuk kombinasi pembebanan (*load combinations)* sesuai faktor skala yang disyaratkan standar desain.

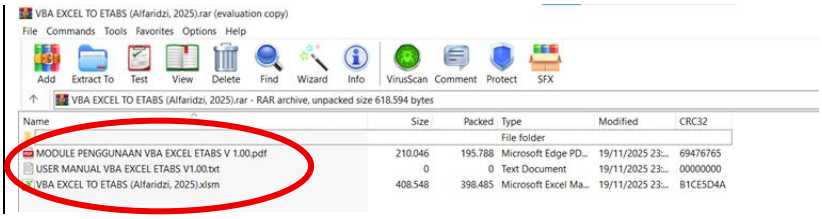
4.4 Algoritma Ekstraksi Hasil

Fungsi ekstraksi data dikembangkan untuk menarik kembali hasil analisis krusial dari basis data ETABS ke dalam lembar kerja Excel. Logika ini memfilter data output untuk mendapatkan nilai maksimum dari reaksi perletakan, gaya elemen, dan simpangan struktur. Hasil ekstraksi disajikan dalam format tabel terstruktur yang siap digunakan untuk evaluasi, menghilangkan risiko kesalahan dari program serta human error.

5 Panduan Penggunaan

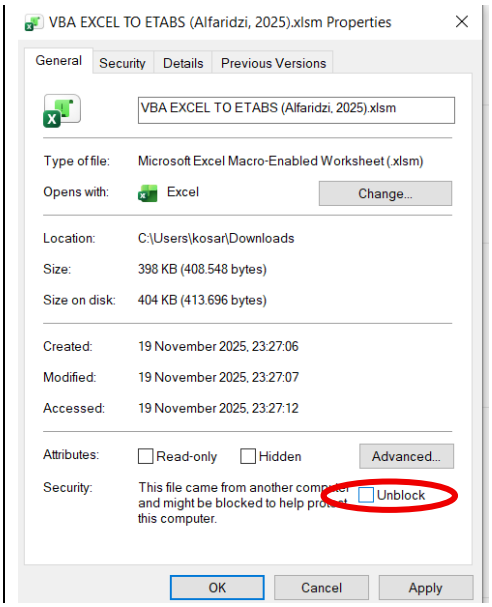
5.1 Extract File Excel dari ZIP

Sebelum bisa menggunakan program, pengguna mengunduh berkas terlebih dahulu, lalu melakukan extract semua file yang ada



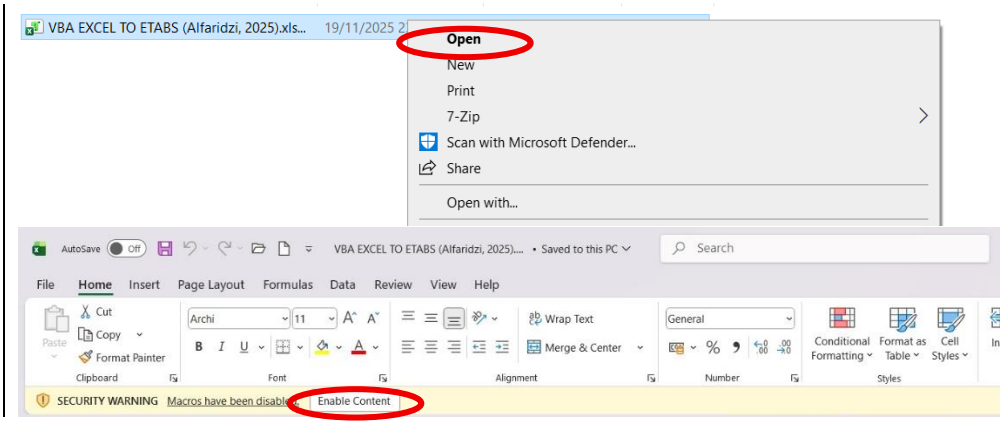
5.2 Unblock file Excel Macro

Selanjutnya klik kanan pada berkas, lalu pilih Properties. Beri tanda centang pada opsi “Unblock”, kemudian klik Apply dan OK.



5.3 Enable Macro pada Excel

Klik dua kali berkas .xlsm untuk membukanya, atau klik kanan berkas lalu pilih opsi "Open". Jika muncul peringatan "Security Warning", klik "Enable Content" agar makro dapat berjalan. Setelah itu, laman menu utama akan muncul secara otomatis.



5.4 Laman Menu Utama

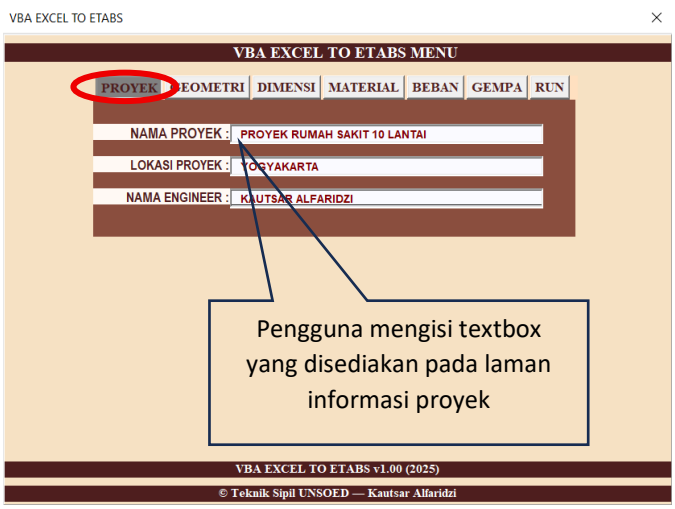
Laman menu utama terdiri atas:

1) Menu data proyek
2) Menu data geometri struktur
3) Menu dimensi struktur
4) Menu material dan konfigurasi
5) Menu pembebanan
6) Menu parameter gempa
7) Menu run

5.5 Laman Informasi Proyek

Pada laman Informasi proyek, pengguna memasukkan data sebagai berikut:

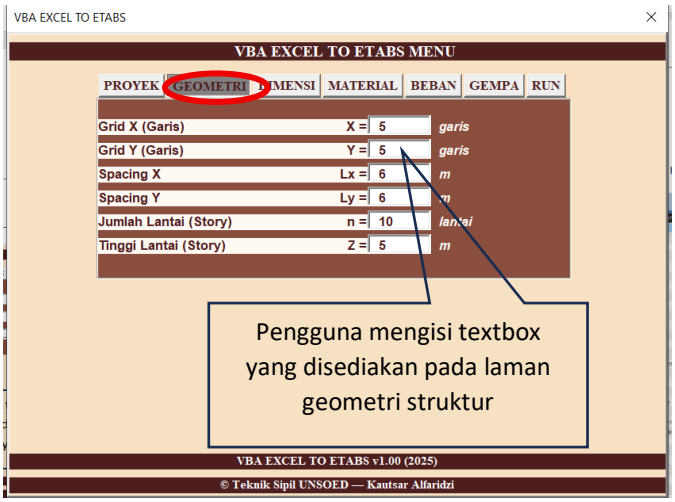
1) Nama proyek yang dikerjakan
2) Lokasi proyek yang dikerjakan
3) Nama pengguna yang mengerjakan



5.6 Laman Data Geometri Struktur

Pada laman geometri struktur, pengguna memasukkan data sebagai berikut:

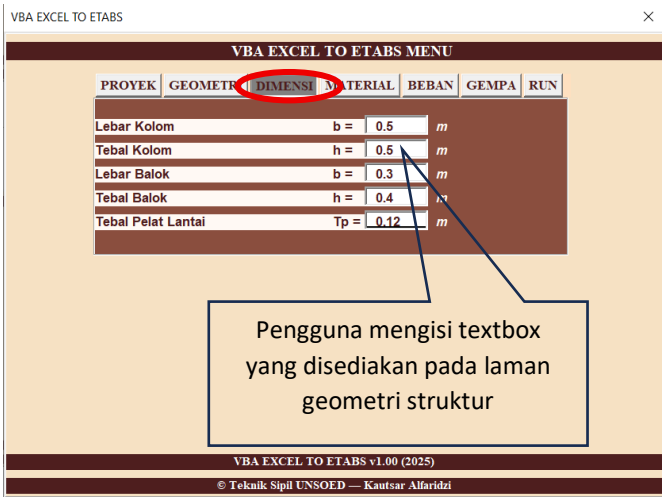
1) Jumlah grid X yang direncanakan
2) Jumlah grid Y yang direncanakan
3) Panjang bentang X yang direncanakan
4) Panjang bentang Y yang direncanakan
5) Jumlah lantai yang direncanakan
6) Ketinggian tiap lantai yang direncanakan



5.7 Laman Dimensi Struktur

Pada laman dimensi struktur, pengguna memasukkan data sebagai berikut:

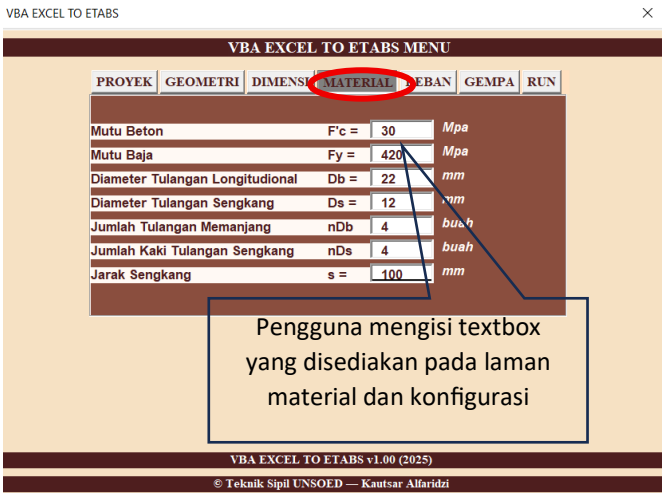
1) Lebar dari kolom yang direncanakan
2) Tebal dari kolom yang direncanakan
3) Lebar dari balok yang direncanakan
4) Tebal dari balok yang direncanakan
5) Tebal pelat lantai yang direncanakan



5.8 Laman Material dan Konfigurasi

Pada material dan konfigurasi, pengguna memasukkan data sebagai berikut:

1) Mutu beton pada setiap elemen struktur yang direncanakan
2) Mutu baja pada setiap elemen struktur yang direncanakan
3) Diameter tulangan longitudional yang direncanakan

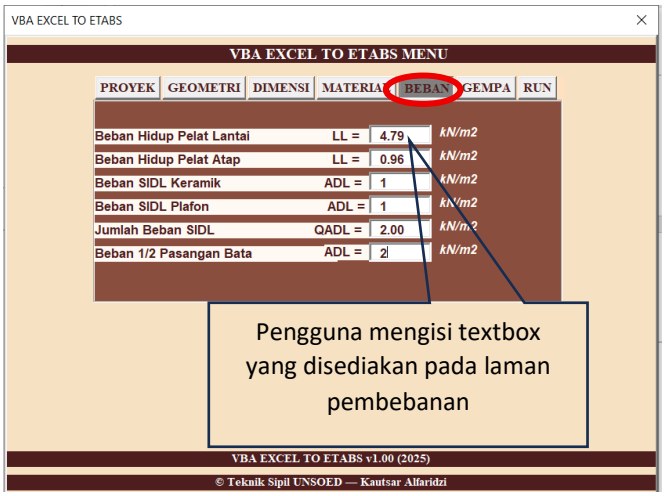


4) Diameter tulangan sengkang yang direncanakan
5) Jumlah tulangan longitudional yang direncanakan
6) Jumlah tulangan sengkang yang direncanakan
7) Jarak sengkang yang direncanakan

5.9 Laman Pembebanan

Pada laman pembebanan, pengguna memasukkan data sebagai berikut:

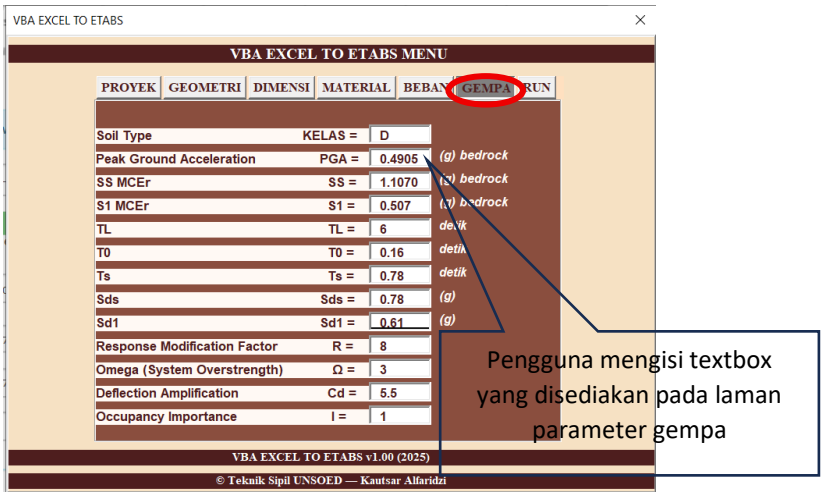
1) Beban hidup pada pelat lantai yang direncanakan
2) Beban hidup pada pelat atap yang direncanakan
3) Beban mati tambah keramik
4) Beban mati tambah plafon
5) Beban mati tambah ½ pasangan bata



5.10 Laman Parameter Gempa

Pada laman parameter gempa, pengguna memasukkan data sebagai berikut:

1) Kelas tanah (soil type)
2) Peak ground acceleration (PGA)
3) Percepatan spectral periode pendek (SS MCEr)
4) Percepatan spectral periode 1 detik (S1 MCEr)
5) Periode transisi panjang (TL)
6) Periode awal (T0)
7) Periode pendek (Ts)
8) Percepatan spektral desain periode pendek (Sds)
9) Percepatan spektral periode 1 detik (Sd1)
10) Response modification factor (R)
11) System overstrength (Ω)
12) Deflection amplification



5.11 Laman Run

Pada laman Run, pengguna melakukan seleksi data melalui pengaktifan checkbox sebagai berikut:

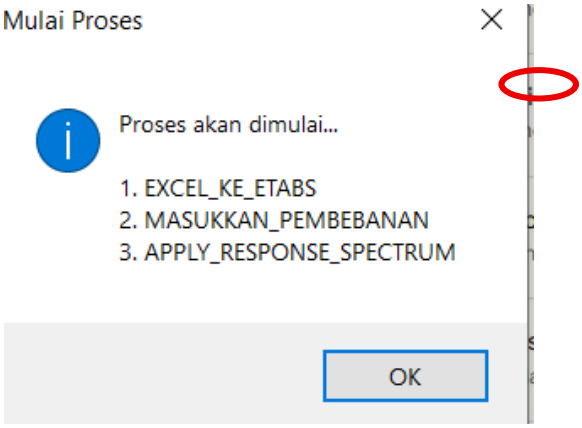
1) Checkbox create model ETABS
2) Checkbox masukkan pembebanan
3) Checkbox apply response spectrum
4) Checkbox run and design check
5) Checkbox export joint reaction
6) Checkbox base reaction
7) Checkbox export beam forces
8) Checkbox export column forces

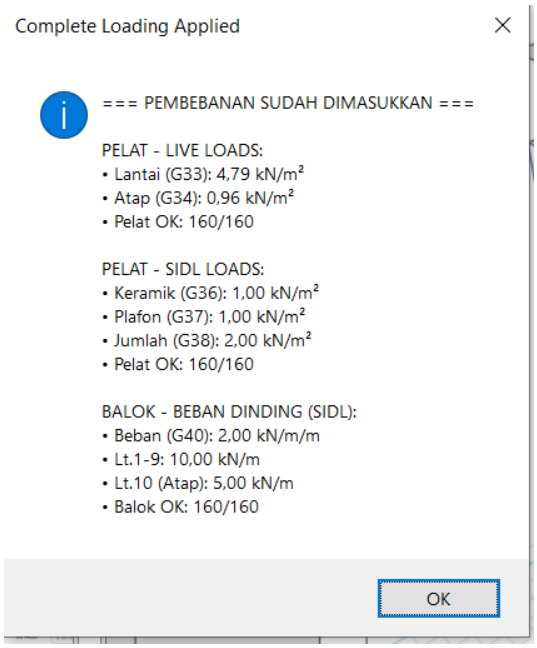
9) Checkbox export node displacement

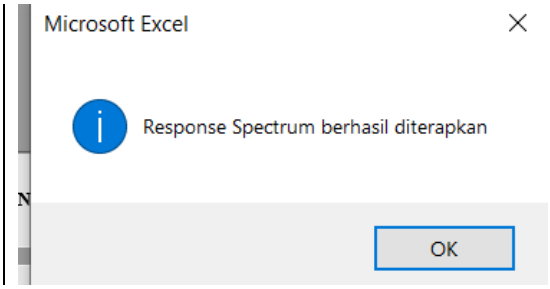
5.12 Proses Analisis

Setelah melakukan seleksi checkbox pada laman run, maka program akan melakukan analisis dengan urutan proses berikut:

1) Konversi data dari Laman sebelumnya ke ETABS (Excel_Ke_ETABS).
2) Perhitungan serta penentuan pembebanan struktur berdasarkan input laman pembebanan (Masukkan Pembebanan).
3) Penerapan beban pada model struktur (Complete Loading Applied).
4) Pembuatan serta penerapan response spectrum untuk analisis dinamis linier berdasarkan laman parameter gempa (Apply_Response_Spectrum).
5) Verifikasi kekuatan elemen struktur melalui design check yang terotomatisasi.



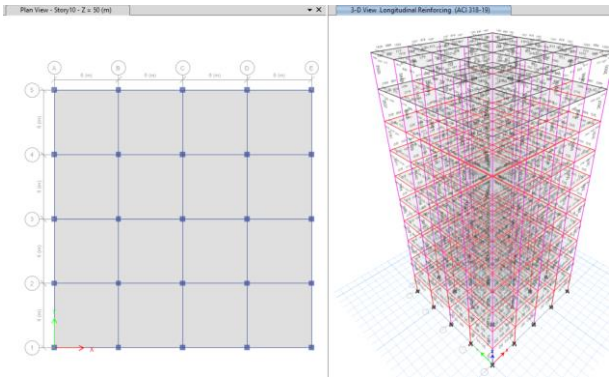


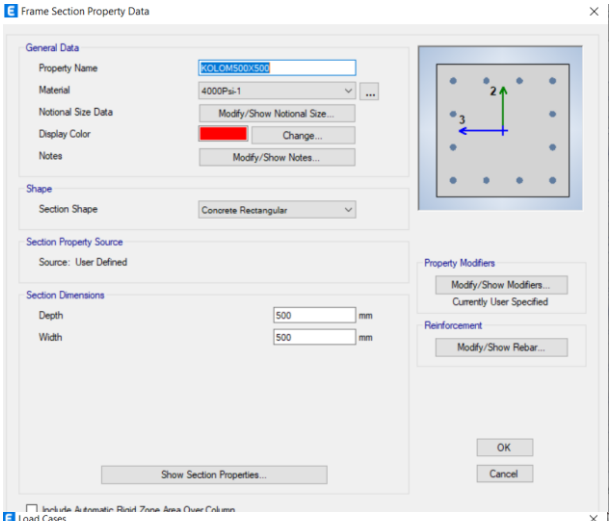


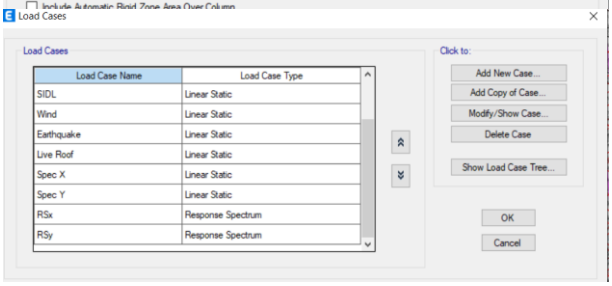
5.13 Output

Setelah sistem sudah melakukan running maka akan keluar hasil analisis struktur yang komprehensif dan detail. Output yang dihasilkan mencakup berbagai jenis informasi mulai dari pemodelan, hingga hasil analisis dan design check. Karena volume output yang cukup besar, pada sub bab ini hanya menampilkan beberapa contoh output yang relevan. Beberapa contoh output yang dihasilkan oleh proses analisis adalah sebagai berikut:

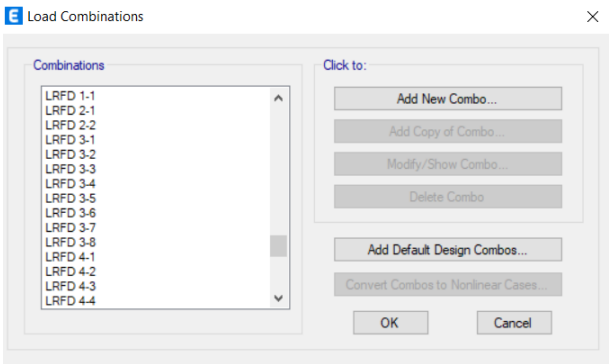
1) Pemodelan struktur bangunan yang menampilkan geometri lengkap rangka struktur beton bertulang.
2) Daftar load cases yang berisi seluruh kombinasi pembebanan statis dan dinamis yang didefinisikan dalam sistem.
3) Data properti penampang elemen struktur yang mencakup informasi material dan dimensi penampang sesuai dengan input.
4) Kombinasi pembebanan (Load Combinations) sesuai dengan standar desain yang digunakan (LRFD dan ASD)

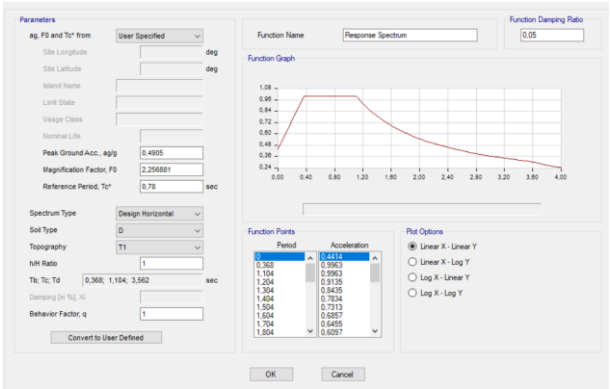






5) Parameter spektrum respons gempa yang ditampilkan dalam bentuk tabel dan grafik
6) Hasil ekstraksi analisis reaksi tumpuan (Joint reaction dan Base Reaction) pada setiap lokasi tumpuan struktur.





	A	B	C	D	E	F	G	H	I	J	K
1	Label	Joint ID	Case/Combo	Type	FX (kN)	FY (kN)	FZ (kN)	MX (kNm)	MY (kNm)	MZ (kNm)	
152	1	1	LRFD 1-1	Load Combo	10,8151907	10,8151907	1963,578399	-17,8436226	17,8436226	-1,7089E-15	
153	2	3	LRFD 1-1	Load Combo	-0,685640575	13,04213012	2703,243536	-21,51777571	-1,131215528	-1,7089E-15	
154	3	5	LRFD 1-1	Load Combo	2,88222E-14	12,62281227	2673,761583	-20,82595718	2,99815E-14	-1,7089E-15	
155	4	7	LRFD 1-1	Load Combo	0,685640575	13,04213012	2703,243536	-21,51777571	1,131215528	-1,7089E-15	
156	5	9	LRFD 1-1	Load Combo	-10,8151907	10,8151907	1963,578399	-17,8436226	-17,8436226	-1,7089E-15	
157	6	11	LRFD 1-1	Load Combo	13,04213012	-0,685640575	2703,243536	1,131215528	21,51777571	-1,7089E-15	
158	7	13	LRFD 1-1	Load Combo	-1,437220583	-1,437220583	3340,739335	2,371222329	-2,371222329	-1,7089E-15	
159	8	15	LRFD 1-1	Load Combo	-3,83023E-14	-1,242890845	3274,906391	2,050604173	-7,66041E-14	-1,7089E-15	
160	9	17	LRFD 1-1	Load Combo	1,437220583	-1,437220583	3340,739335	2,371222329	2,371222329	-1,7089E-15	
161	10	19	LRFD 1-1	Load Combo	-13,04213012	-0,685640575	2703,243536	1,131215528	-21,51777571	-1,7089E-15	
162	11	21	LRFD 1-1	Load Combo	12,62281227	-5,21533E-14	2673,761583	9,87421E-14	20,82595718	-1,7089E-15	
163	12	23	LRFD 1-1	Load Combo	-1,242890845	-2,21027E-15	3274,906391	2,0333E-14	-2,050604173	-1,7089E-15	
164	13	25	LRFD 1-1	Load Combo	7,07836E-14	1,26443E-14	3222,860822	-9,17788E-17	1,07441E-13	-1,7089E-15	
165	14	27	LRFD 1-1	Load Combo	1,242890845	-7,48093E-14	3274,906391	1,48278E-13	2,050604173	-1,7089E-15	
166	15	29	LRFD 1-1	Load Combo	-12,62281227	-8,50501E-15	2673,761583	4,28752E-14	-20,82595718	-1,7089E-15	
167	16	31	LRFD 1-1	Load Combo	13,04213012	0,685640575	2703,243536	-1,131215528	21,51777571	-1,7089E-15	
168	17	33	LRFD 1-1	Load Combo	-1,437220583	1,437220583	3340,739335	-2,371222329	-2,371222329	-1,7089E-15	
169	18	35	LRFD 1-1	Load Combo	3,39449E-14	1,242890845	3274,906391	-2,050604173	5,073E-14	-1,7089E-15	
170	19	37	LRFD 1-1	Load Combo	1,437220583	1,437220583	3340,739335	-2,371222329	2,371222329	-1,7089E-15	
171	20	39	LRFD 1-1	Load Combo	-13,04213012	0,685640575	2703,243536	-1,131215528	-21,51777571	-1,7089E-15	
172	21	41	LRFD 1-1	Load Combo	10,8151907	-10,8151907	1963,578399	17,8436226	17,8436226	-1,7089E-15	
173	22	43	LRFD 1-1	Load Combo	-0,685640575	-13,04213012	2703,243536	21,51777571	-1,131215528	-1,7089E-15	
174	23	45	LRFD 1-1	Load Combo	2,89536E-14	-12,62281227	2673,761583	20,82595718	4,66559E-14	-1,7089E-15	
175	24	47	LRFD 1-1	Load Combo	0,685640575	-13,04213012	2703,243536	21,51777571	1,131215528	-1,7089E-15	
176	25	49	LRFD 1-1	Load Combo	-10,8151907	-10,8151907	1963,578399	17,8436226	-17,8436226	-1,7089E-15	
177	1	1	LRFD 2-1	Load Combo	14,44833796	14,44833796	2265,396196	-23,83783115	23,83783115	-2,4459E-15	
178	2	3	LRFD 2-1	Load Combo	-1,172717501	24,33234703	3574,489235	-40,14512824	-1,934827512	-2,4459E-15	
179	3	5	LRFD 2-1	Load Combo	4,79608E-14	23,56385087	3523,63032	-38,87721204	5,86153E-14	-2,4459E-15	
180	4	7	LRFD 2-1	Load Combo	1,172717501	24,33234703	3574,489235	-40,14512824	1,934827512	-2,4459E-15	

DATA INPUT | JOINT REACTION | BASE REACTION

1	Label	Case/Combo	Type	FX (kN)	FY (kN)	FZ (kN)	MX (kNm)	MY (kNm)
2	1	Dead	Load Case	-5,59339E-14	1,18234E-14	29836,5371	358038,4452	-358038,4452
3	2	Live	Load Case	-3,5925E-14	-9,09495E-15	25384,32	304611,84	-304611,84
4	3	Modal	Load Case	0,055684966	0,022507643	4,26326E-17	-0,764306897	1,890930962
5	4	SIDL	Load Case	-5,00222E-14	-2,45564E-14	20064	240768	-240768
6	5	Wind	Load Case	0	0	0	0	0
7	6	Earthquake	Load Case	0	0	0	0	0
8	7	Live Roof	Load Case	0	0	0	0	0
9	8	Spec X	Load Case	-4460,59917	-7,32456E-08	5,3551E-12	3,33973E-06	-169358,7362
10	9	Spec Y	Load Case	-7,90603E-08	-4460,59917	1,62981E-12	169358,7362	-2,77441E-06
11	10	RSx	Load Case	0	0	0	0	0
12	11	RSy	Load Case	0	0	0	0	0
13	12	~StaticRSx	Load Case	0	0	0	0	0
14	13	~StaticRSy	Load Case	0	0	0	0	0
15	14	LRFD 1-1	Load Combo	-1,48339E-13	-1,78261E-14	69860,75194	838329,0233	-838329,0233
16	15	LRFD 2-1	Load Combo	-1,84627E-13	-2,98314E-14	100495,5565	1205946,678	-1205946,678
17	16	LRFD 2-2	Load Combo	-1,84627E-13	-2,98314E-14	100495,5565	1205946,678	-1205946,678
18	17	LRFD 3-1	Load Combo	-1,63072E-13	-2,43745E-14	85264,96452	1023179,574	-1023179,574
19	18	LRFD 3-2	Load Combo	-1,84627E-13	-2,98314E-14	100495,5565	1205946,678	-1205946,678
20	19	LRFD 3-3	Load Combo	-1,27147E-13	-1,52795E-14	59880,64452	718567,7342	-718567,7342
21	20	LRFD 3-4	Load Combo	-1,27147E-13	-1,52795E-14	59880,64452	718567,7342	-718567,7342
22	21	LRFD 3-5	Load Combo	-1,63072E-13	-2,43745E-14	85264,96452	1023179,574	-1023179,574
23	22	LRFD 3-6	Load Combo	-1,84627E-13	-2,98314E-14	100495,5565	1205946,678	-1205946,678
24	23	LRFD 3-7	Load Combo	-1,27147E-13	-1,52795E-14	59880,64452	718567,7342	-718567,7342
25	24	LRFD 3-8	Load Combo	-1,27147E-13	-1,52795E-14	59880,64452	718567,7342	-718567,7342
26	25	LRFD 4-1	Load Combo	-1,63072E-13	-2,43745E-14	85264,96452	1023179,574	-1023179,574
27	26	LRFD 4-2	Load Combo	-1,63072E-13	-2,43745E-14	85264,96452	1023179,574	-1023179,574
28	27	LRFD 4-3	Load Combo	-1,63072E-13	-2,43745E-14	85264,96452	1023179,574	-1023179,574
29	28	LRFD 4-4	Load Combo	-1,63072E-13	-2,43745E-14	85264,96452	1023179,574	-1023179,574
30	29	LRFD 4-5	Load Combo	-1,63072E-13	-2,43745E-14	85264,96452	1023179,574	-1023179,574

DATA INPUT | JOINT REACTION | BASE REACTION

6 Penutup

Bab ini memuat Kesimpulan akhir dari seluruh rangkain pengembangan program STRUKXEL, serta saran-saran strategis untuk pemanfaaatan dan pengembangan sistem di masa mendatang.

6.1 Kesimpulan

Berdasarkan seluruh tahapan pengembangan dan pengujian yang telah dilakukan, program komputer STRUKXEL terbukti berhasil berfungsi sebagai solusi otomasi desain struktur beton bertulang yang efektif. Pencapaian kinerja program dapat disimpulkan sebagai berikut:

a) Efisiensi kerja signifikan, program mampu mereduksi langkah kerja desain sebesar 88%, menyederhanakan proses manual yang semula membutuhkan 100 langkah menjadi hanya 12 langkah otomatis, sehingga mempercepat waktu penyelesaian proyek secara drastis .
b) Akurasi tinggi, hasil validasi menunjukkan tingkat kesesuaian yang sangat tinggi dengan pemodelan manual, mencatatkan rata-rata deviasi (*error*) hanya sebesar 0,057% pada parameter kunci seperti gaya geser dasar dan simpangan lantai. Hal ini membuktikan keandalan algoritma yang diterapkan .
c) Eliminasi kesalahan manusia, sistem berhasil menghilangkan potensi kesalahan umum akibat campur tangan manusia (*human error*) pada tahap input koordinat, definisi properti, dan aplikasi beban, sehingga menjamin integritas dan konsisten data.

6.2 Saran Pemanfaatn dan Pengembangan

STRUKXEL direkomendasikan untuk digunakan oleh praktisi rekayasa sipil sebagai alat bantu desain awal (*preliminary design*) yang cepat dan terukur, serta sebagai media pembelajaran di lingkungan akademik. Untuk pengembangan selanjutnya, disarankan memperluas kemampuan program agar dapat menangani struktur dengan ketidakberaturan geometri serta integrasi dengan platform *Building Information Modeling* (BIM) guna mendukung digitalisasi konstruksi yang lebih menyeluruh .

7 Referensi

Andrés David Torres Carranza. 2024. Development of an application implementing the ETABS and VBA Excel API, for the automated analysis of structural irregularities taking into account NSR - 10. *Escuela Colombiana de Ingeniería Julio Garavito.* 15(1): 37–48.

Aslay, S.E.; Dede, T. & Kaveh, A. 2024. Integrated Design Optimization Process for Building Projects. *Periodica Polytechnica Civil Engineering* 68(4): 1175–1183, https://doi.org/10.3311/PPci.37113.

B. Anjaneyulu, K.J.P. 2020. Analysis and Design of Flat Slab. *International journal of Latest Trends in Engineering & Technology* 7(1): 105–112, https://doi.org/10.21172/1.71.018.

Badan Standardisasi Nasional Indonesia. 2019. Persyaratan Beton Struktural untuk Bangunan Gedung dan Penjelasan (SNI 2847:2019). *Standar Nasional Indonesia (SNI)* (8): 653–659.

Badan Standardisasi Nasional Indonesia. 2020a. Penetapan Standar Nasional Indonesia 1727 : 2020 Beban Desain Minimum dan Kriteria Terkait Untuk Bangunan Gedung dan Struktur. *Badan Standarisasi Nasional 1727:2020* (8): 1–336.

Badan Standardisasi Nasional Indonesia, 2020. 2020b. Standar Nasional Indonesia (SNI 1729:2019) tentang Perencanaan Ketahanan Gempa untuk Struktur Bangunan. (8).

Cormos, R.; Neagoe, C.A.; Ciolca, M. & Hadar, A. 2023. VBA fatigue analysis program for metallic structural components preliminary design. *Incas Bulletin* 15(4): 93–106, https://doi.org/10.13111/2066-8201.2023.15.4.8.

Johnson, L. 2023. Automating Structural Analysis with VBA in Excel. *International Journal of Engineering Research and Applications* 12(4)(456–462).

Jorge L. P. Tamayo, Vitor C. Mucillo, B.G.B. 2023. Integration of Excel VBA with professional software for the structural analysis and design of civil structures. *Computer Applications in Engineering Education*, https://doi.org/10.1002/cae.22607.

PuSGeN, K.P. 2021. Aplikasi Desain Spektra Indonesia 2021.

Quansah, A.; Xiao, Z. & Ntaryamira, T. 2017. Parametric Investigation of Plane Frame Structure Using Excel Vba And Sap2000 oapi Based Programming. *International Journal of Engineering Research and Applications* 07(07): 14–23, https://doi.org/10.9790/9622-0707091423.

Tampubolon, S.P.; Fauzan, A.S.; Frans, P.L.; Gaspersz, W.; Oktavia, Deffa Diana & Roesdiana, T.; Vendry, P.; Soselisa, J.F.; Frans, P.L.; Hutubessy, V.R.R.; Widyatama &Suprapty; Kiriw, V.P. & Badan Standardisasi Nasional. 2022. *1983 Peraturan-Pembebanan-Indonesia*. *Jurnal konstruksi dan infrastruktur* 8, pp. 1–126, 2022.

Lampiran

Lampiran File VBA Excel ke ETABS (qr code)

https://drive.google.com/drive/folders/1s53HTpG8G9ileW5gDJSOD28wCo9P5c6K?usp=sharing

Lampiran 1 Module1 (EXCEL_KE_ETABS)

```
Option Explicit

Const eUnits_kg_m_C As Long = 6

Public Sub EXCEL_KE_ETABS()
    On Error GoTo ErrHandler
    Application.ScreenUpdating = False
    Application.Calculation = xlCalculationManual
    Application.StatusBar = "Connecting to ETABS..."

    Dim EtabsObject As Object, SapModel As Object, ret As Long
    On Error Resume Next
    Dim myHelper As Object: Set myHelper =
CreateObject("ETABSv1.Helper")
    If Not myHelper Is Nothing Then
        Set EtabsObject =
myHelper.GetObject("CSI.ETABS.API.ETABSObject")
        If EtabsObject Is Nothing Then Set EtabsObject =
myHelper.CreateObjectProgID("CSI.ETABS.API.ETABSObject")
    End If
    If EtabsObject Is Nothing Then Set EtabsObject =
CreateObject("CSI.ETABS.API.ETABsObject")
    On Error GoTo ErrHandler

    ret = EtabsObject.ApplicationStart()
    Set SapModel = EtabsObject.SapModel

    Dim ws As Worksheet
    Set ws = ThisWorkbook.Sheets("DATA INPUT")
```

```

    Dim nStory As Long, h As Double, nGX As Long, nGY As Long, SpX As
Double, SpY As Double
    nStory = CLng(val(ws.Range("G12").Value))
    h = CDbl(val(ws.Range("G13").Value))
    nGX = CLng(val(ws.Range("G8").Value))
    nGY = CLng(val(ws.Range("G9").Value))
    SpX = CDbl(val(ws.Range("G10").Value))
    SpY = CDbl(val(ws.Range("G11").Value))

    Dim columnWidth As Double, columnDepth As Double, columnCover As
Double
    Dim beamWidth As Double, beamDepth As Double, beamCover As Double
    Dim slabThickness As Double
    Dim mainBarDia As Double, tieBarDia As Double
    Dim numLongBars As Long, numTieLegs As Long, tieSpacing As Double
    columnWidth = CDbl(ws.Range("G20").Value)
    columnDepth = CDbl(ws.Range("G21").Value)
    beamWidth = CDbl(ws.Range("G22").Value)
    beamDepth = CDbl(ws.Range("G23").Value)
    slabThickness = CDbl(ws.Range("G24").Value)
    mainBarDia = CDbl(ws.Range("G25").Value)
    tieBarDia = CDbl(ws.Range("G26").Value)
    numLongBars = CLng(val(ws.Range("G27").Value))
    numTieLegs = CLng(val(ws.Range("G28").Value))
    tieSpacing = CDbl(val(ws.Range("G29").Value))
    columnCover = 0.04
    beamCover = 0.04

    Dim fcValue As Double: fcValue = CDbl(ws.Range("G17").Value)
    Dim fyValue As Double: fyValue = CDbl(ws.Range("G18").Value)

    If nStory < 1 Or nGX < 2 Or nGY < 2 Or h <= 0 Or SpX <= 0 Or SpY
<= 0 Then Err.Raise vbObjectError + 100, , "Input grid tidak valid.
Cek C2,C12,C5,C6,C13,C14."
    If columnWidth <= 0 Or columnDepth <= 0 Or beamWidth <= 0 Or
beamDepth <= 0 Or slabThickness <= 0 Then Err.Raise vbObjectError +
101, , "Input dimensi tidak valid. Cek F4,H4,F5,H5,F6."
    If fcValue <= 0 Then Err.Raise vbObjectError + 102, , "Input kuat
tekan beton tidak valid. Cek F7."
    If fyValue <= 0 Then Err.Raise vbObjectError + 103, , "Input kuat
leleh baja tidak valid. Cek F8."

    ret = SapModel.InitializeNewModel(eUnits_kg_m_C)
    ret = SapModel.SetPresentUnits(eUnits_kg_m_C)
    ret = SapModel.DesignConcrete.SetCode("ACI 318-19")
    ret = SapModel.DesignSteel.SetCode("AISC 360-22")
    ret = SapModel.File.NewGridOnly(nStory, h, h, nGX, nGY, SpX, SpY)

    Dim concreteGrade As String
    concreteGrade = "BETON FC " & Format(fcValue, "0") & " (G17)"

    ret = SapModel.PropMaterial.AddMaterial(concreteGrade, 2, "", "",
"Custom concrete from Excel cell F7")

    Dim density As Double, modulusOfElasticity As Double, poissonRatio
As Double, thermalExpansion As Double
    density = 2.4
```

```
    modulusOfElasticity = 4700 * Sqr(fcValue)
    poissonRatio = 0.2
    thermalExpansion = 0.0000099

    Dim fcInKNPerM2 As Double: fcInKNPerM2 = fcValue * 1000
    ret = SapModel.PropMaterial.SetMPIsotropic(concreteGrade,
modulusOfElasticity * 1000, poissonRatio, thermalExpansion)
    ret = SapModel.PropMaterial.SetWeightAndMass(concreteGrade, 2,
density)

    On Error Resume Next
    ret = SapModel.PropMaterial.SetOConcrete_1(concreteGrade,
fcInKNPerM2, False, 0, 0, 0, 0, 0, 0)
    If ret <> 0 Then
        ret = SapModel.PropMaterial.SetOConcrete_1(concreteGrade,
fcInKNPerM2, False)
        If ret <> 0 Then
            ret = SapModel.PropMaterial.SetOConcrete(concreteGrade,
fcInKNPerM2, False, 0, 0, 0, 0)
        End If
    End If
    On Error GoTo ErrHandler

    Dim fuValue As Double, fyeValue As Double, fueValue As Double
    fuValue = 1.25 * fyValue
    fyeValue = 1.1 * fyValue
    fueValue = 1.1 * fuValue

    Dim steelGrade As String
    steelGrade = "BAJA TULANGAN FY " & Format(fyValue, "0") & " (G18)"

    ret = SapModel.PropMaterial.AddMaterial(steelGrade, 6, "", "",
"Rebar from Excel F8")

    Dim steelDensity As Double, steelE As Double, steelNu As Double,
steelAlpha As Double
    steelDensity = 7.85
    steelE = 200000
    steelNu = 0.3
    steelAlpha = 0.000012

    Dim fy_kN As Double, fu_kN As Double, fye_kN As Double, fue_kN As
Double
    fy_kN = fyValue * 1000
    fu_kN = fuValue * 1000
    fye_kN = fyeValue * 1000
    fue_kN = fueValue * 1000

    ret = SapModel.PropMaterial.SetMPIsotropic(steelGrade, steelE *
1000, steelNu, steelAlpha)
    ret = SapModel.PropMaterial.SetWeightAndMass(steelGrade, 2,
steelDensity)

    On Error Resume Next
    ret = SapModel.PropMaterial.SetORebar_1(steelGrade, fy_kN, fu_kN,
fye_kN, fue_kN, 0, 0, 0, 0, 0, False)
    If ret <> 0 Then
```

```
        ret = SapModel.PropMaterial.SetORebar(steelGrade, fy_kN,
fu_kN, fye_kN, fue_kN, 0, 0, 0, 0, False)
    End If
    On Error GoTo ErrHandler

    Dim beamSection As String, columnSection As String, slabSection As
String
    beamSection = "BALOK" & CInt(beamWidth * 1000) & "X" &
CInt(beamDepth * 1000)
    columnSection = "KOLOM" & CInt(columnWidth * 1000) & "X" &
CInt(columnDepth * 1000)
    slabSection = "PELAT " & CInt(slabThickness * 1000)

    On Error Resume Next
    SapModel.PropFrame.Delete beamSection
    SapModel.PropFrame.Delete columnSection
    SapModel.PropArea.Delete slabSection
    On Error GoTo ErrHandler

    ret = SapModel.PropFrame.SetRectangle(beamSection, concreteGrade,
beamDepth, beamWidth)
    ret = SapModel.PropFrame.SetRebarBeam(beamSection, steelGrade,
steelGrade, beamCover, beamCover, 0, 0, 0, 0)

    ret = SapModel.PropFrame.SetRectangle(columnSection,
concreteGrade, columnDepth, columnWidth)
    ret = SapModel.PropFrame.SetRebarColumn(columnSection, steelGrade,
steelGrade, 1, 1, columnCover, 0, numLongBars, numLongBars,
CStr(CInt(mainBarDia)), CStr(CInt(tieBarDia)), tieSpacing * 0.001,
numTieLegs, numTieLegs, True)

    ret = SapModel.PropArea.SetSlab(slabSection, 0, 1, concreteGrade,
slabThickness)

    Dim i As Integer, j As Integer, k As Integer
    Dim x1 As Double, y1 As Double, z1 As Double
    Dim x2 As Double, y2 As Double, z2 As Double

    x1 = 0: y1 = 0: z1 = 0
    x2 = 0: y2 = 0: z2 = h
    For i = 0 To nStory - 1
        y1 = 0: y2 = 0
        For k = 0 To nGY - 1
            x1 = 0: x2 = 0
            For j = 0 To nGX - 1
                SapModel.FrameObj.AddByCoord x1, y1, z1, x2, y2, z2,
"", columnSection
                x1 = x1 + SpX: x2 = x2 + SpX
            Next j
            y1 = y1 + SpY: y2 = y2 + SpY
        Next k
        z1 = z1 + h: z2 = z2 + h
    Next i

    x1 = 0: y1 = 0: z1 = h
    x2 = SpX: y2 = 0: z2 = h
    For i = 0 To nStory - 1
        y1 = 0: y2 = 0
```

```
        For k = 0 To nGY - 1
            x1 = 0: x2 = SpX
            For j = 0 To nGX - 2
                SapModel.FrameObj.AddByCoord x1, y1, z1, x2, y2, z2,
"", beamSection
                x1 = x1 + SpX: x2 = x2 + SpX
            Next j
            y1 = y1 + SpY: y2 = y2 + SpY
        Next k
        z1 = z1 + h: z2 = z2 + h
    Next i

    x1 = 0: y1 = 0: z1 = h
    x2 = 0: y2 = SpY: z2 = h
    For i = 0 To nStory - 1
        y1 = 0: y2 = SpY
        For k = 0 To nGY - 2
            x1 = 0: x2 = 0
            For j = 0 To nGX - 1
                SapModel.FrameObj.AddByCoord x1, y1, z1, x2, y2, z2,
"", beamSection
                x1 = x1 + SpX: x2 = x2 + SpX
            Next j
            y1 = y1 + SpY: y2 = y2 + SpY
        Next k
        z1 = z1 + h: z2 = z2 + h
    Next i

    Dim X() As Double, Y() As Double, z() As Double
    Dim diaphragmName As String, areaName As String
    ReDim X(3), Y(3), z(3)
    X(0) = 0: Y(0) = 0
    X(1) = SpX: Y(1) = 0
    X(2) = SpX: Y(2) = SpY
    X(3) = 0: Y(3) = SpY

    For i = 0 To nStory - 1
        z(0) = h * (i + 1): z(1) = h * (i + 1): z(2) = h * (i + 1):
z(3) = h * (i + 1)
        diaphragmName = "D" & (i + 1)
        On Error Resume Next
        SapModel.Diaphragm.Delete diaphragmName
        On Error GoTo ErrHandler
        ret = SapModel.Diaphragm.SetDiaphragm(diaphragmName, 0)

        Y(0) = 0: Y(1) = 0: Y(2) = SpY: Y(3) = SpY
        For k = 0 To nGY - 2
            X(0) = 0: X(1) = SpX: X(2) = SpX: X(3) = 0
            For j = 0 To nGX - 2
                SapModel.AreaObj.AddByCoord 4, X, Y, z, areaName,
slabSection
                ret = SapModel.AreaObj.SetDiaphragm(areaName,
diaphragmName)
                X(0) = X(0) + SpX: X(1) = X(1) + SpX: X(2) = X(2) +
SpX: X(3) = X(3) + SpX
            Next j
            Y(0) = Y(0) + SpY: Y(1) = Y(1) + SpY: Y(2) = Y(2) + SpY:
Y(3) = Y(3) + SpY
```

```
261             Next k
262         Next i
263
264         Dim restrains() As Boolean
265         ReDim restrains(5)
266         For i = 0 To 5: restrains(i) = True: Next i
267         Dim totalSupports As Integer: totalSupports = nGX * nGY
268         For i = 0 To totalSupports - 1
269             SapModel.PointObj.SetRestraint i * 2 + 1, restrains
270         Next i
271
272         Application.StatusBar = "Setting Property Modifiers..."
273
274         Dim beamModifiers() As Double
275         ReDim beamModifiers(7)
276         For i = 0 To 7
277             beamModifiers(i) = 1
278         Next i
279         beamModifiers(3) = 0.01    '
280         beamModifiers(4) = 0.35
281         beamModifiers(5) = 0.35
282         ret = SapModel.PropFrame.SetModifiers(beamSection, beamModifiers)
283
284         ' Modifier Kolom
285         Dim columnModifiers() As Double
286         ReDim columnModifiers(7)
287         For i = 0 To 7
288             columnModifiers(i) = 1
289         Next i
290         columnModifiers(4) = 0.7
291         columnModifiers(5) = 0.7
292         ret = SapModel.PropFrame.SetModifiers(columnSection,
293     columnModifiers)
294
295         ' Modifier Pelat Lantai
296         Dim slabModifiers() As Double
297         ReDim slabModifiers(9)
298         For i = 0 To 9
299             slabModifiers(i) = 1
300         Next i
301         slabModifiers(0) = 0.25
302         slabModifiers(1) = 0.25
303         slabModifiers(2) = 0.25
304         slabModifiers(3) = 0.25
305         slabModifiers(4) = 0.25
306         slabModifiers(5) = 0.25
307         ret = SapModel.PropArea.SetModifiers(slabSection, slabModifiers)
308
309         Dim loadPatternName As String, loadPatternType As Long,
310     selfWeightMultiplier As Double
311         loadPatternName = "SIDL": loadPatternType = 2:
312     selfWeightMultiplier = 0
313         SapModel.LoadPatterns.Add loadPatternName, loadPatternType,
314     selfWeightMultiplier
315         loadPatternName = "Wind": loadPatternType = 6:
316     selfWeightMultiplier = 0
317         SapModel.LoadPatterns.Add loadPatternName, loadPatternType,
318     selfWeightMultiplier
```

```
    loadPatternName = "Earthquake": loadPatternType = 5:
selfWeightMultiplier = 0
    SapModel.LoadPatterns.Add loadPatternName, loadPatternType,
selfWeightMultiplier
    loadPatternName = "Live Roof": loadPatternType = 3:
selfWeightMultiplier = 0
    SapModel.LoadPatterns.Add loadPatternName, loadPatternType,
selfWeightMultiplier

    Dim slabId As String, slabLoadPatternName As String
    Dim slabLoad As Double, slabLoadDir As Integer,
slabIsReplaceExisting As Boolean
    slabId = "ALL": slabLoadPatternName = "Live": slabLoad = 2:
slabLoadDir = 10: slabIsReplaceExisting = True
    SapModel.AreaObj.SetLoadUniform slabId, slabLoadPatternName,
slabLoad, slabLoadDir, slabIsReplaceExisting, "Global", 2
    slabId = "ALL": slabLoadPatternName = "SIDL": slabLoad = 1.5:
slabLoadDir = 10: slabIsReplaceExisting = True
    SapModel.AreaObj.SetLoadUniform slabId, slabLoadPatternName,
slabLoad, slabLoadDir, slabIsReplaceExisting, "Global", 2

       Dim comboName As String, comboType As Long, comboLoadType As
Long
    comboType = 0: comboLoadType = 0

    comboName = "LRFD 1-1"
    SapModel.RespCombo.Add comboName, comboType
    SapModel.RespCombo.SetCaseList comboName, comboLoadType, "Dead",
1.4
    SapModel.RespCombo.SetCaseList comboName, comboLoadType, "SIDL",
1.4

    comboName = "LRFD 2-1"
    SapModel.RespCombo.Add comboName, comboType
    SapModel.RespCombo.SetCaseList comboName, comboLoadType, "Dead",
1.2
    SapModel.RespCombo.SetCaseList comboName, comboLoadType, "SIDL",
1.2
    SapModel.RespCombo.SetCaseList comboName, comboLoadType, "Live",
1.6
    SapModel.RespCombo.SetCaseList comboName, comboLoadType, "Live
Roof", 0.5

    comboName = "LRFD 2-2"
    SapModel.RespCombo.Add comboName, comboType
    SapModel.RespCombo.SetCaseList comboName, comboLoadType, "Dead",
1.2
    SapModel.RespCombo.SetCaseList comboName, comboLoadType, "SIDL",
1.2
    SapModel.RespCombo.SetCaseList comboName, comboLoadType, "Live",
1.6
    SapModel.RespCombo.SetCaseList comboName, comboLoadType, "Live
Roof", 0.5

    comboName = "LRFD 3-1"
    SapModel.RespCombo.Add comboName, comboType
    SapModel.RespCombo.SetCaseList comboName, comboLoadType, "Dead",
1.2
```

```
377         SapModel.RespCombo.SetCaseList comboName, comboLoadType, "SIDL",
378     1.2
379         SapModel.RespCombo.SetCaseList comboName, comboLoadType, "Live", 1
380         SapModel.RespCombo.SetCaseList comboName, comboLoadType, "Live
381     Roof", 1.6
382
383         comboName = "LRFD 3-2"
384         SapModel.RespCombo.Add comboName, comboType
385         SapModel.RespCombo.SetCaseList comboName, comboLoadType, "Dead",
386     1.2
387         SapModel.RespCombo.SetCaseList comboName, comboLoadType, "SIDL",
388     1.2
389         SapModel.RespCombo.SetCaseList comboName, comboLoadType, "Live",
390     1.6
391         SapModel.RespCombo.SetCaseList comboName, comboLoadType, "Wind",
392     0.5
393
394         comboName = "LRFD 3-3"
395         SapModel.RespCombo.Add comboName, comboType
396         SapModel.RespCombo.SetCaseList comboName, comboLoadType, "Dead",
397     1.2
398         SapModel.RespCombo.SetCaseList comboName, comboLoadType, "SIDL",
399     1.2
400         SapModel.RespCombo.SetCaseList comboName, comboLoadType, "Live
401     Roof", 1.6
402         SapModel.RespCombo.SetCaseList comboName, comboLoadType, "Wind",
403     0.5
404
405         comboName = "LRFD 3-4"
406         SapModel.RespCombo.Add comboName, comboType
407         SapModel.RespCombo.SetCaseList comboName, comboLoadType, "Dead",
408     1.2
409         SapModel.RespCombo.SetCaseList comboName, comboLoadType, "SIDL",
410     1.2
411         SapModel.RespCombo.SetCaseList comboName, comboLoadType, "Live
412     Roof", 1.6
413         SapModel.RespCombo.SetCaseList comboName, comboLoadType, "Wind",
414     0.375
415
416         comboName = "LRFD 3-5"
417         SapModel.RespCombo.Add comboName, comboType
418         SapModel.RespCombo.SetCaseList comboName, comboLoadType, "Dead",
419     1.2
420         SapModel.RespCombo.SetCaseList comboName, comboLoadType, "SIDL",
421     1.2
422         SapModel.RespCombo.SetCaseList comboName, comboLoadType, "Live", 1
423         SapModel.RespCombo.SetCaseList comboName, comboLoadType, "Live
424     Roof", 1.6
425
426         comboName = "LRFD 3-6"
427         SapModel.RespCombo.Add comboName, comboType
428         SapModel.RespCombo.SetCaseList comboName, comboLoadType, "Dead",
429     1.2
430         SapModel.RespCombo.SetCaseList comboName, comboLoadType, "SIDL",
431     1.2
432         SapModel.RespCombo.SetCaseList comboName, comboLoadType, "Live",
433     1.6
```

```
434         SapModel.RespCombo.SetCaseList comboName, comboLoadType, "Wind",
435     0.5
436
437         comboName = "LRFD 3-7"
438         SapModel.RespCombo.Add comboName, comboType
439         SapModel.RespCombo.SetCaseList comboName, comboLoadType, "Dead",
440     1.2
441         SapModel.RespCombo.SetCaseList comboName, comboLoadType, "SIDL",
442     1.2
443         SapModel.RespCombo.SetCaseList comboName, comboLoadType, "Live
444     Roof", 1.6
445         SapModel.RespCombo.SetCaseList comboName, comboLoadType, "Wind",
446     0.5
447
448         comboName = "LRFD 3-8"
449         SapModel.RespCombo.Add comboName, comboType
450         SapModel.RespCombo.SetCaseList comboName, comboLoadType, "Dead",
451     1.2
452         SapModel.RespCombo.SetCaseList comboName, comboLoadType, "SIDL",
453     1.2
454         SapModel.RespCombo.SetCaseList comboName, comboLoadType, "Live
455     Roof", 1.6
456         SapModel.RespCombo.SetCaseList comboName, comboLoadType, "Wind",
457     0.375
458
459         comboName = "LRFD 4-1"
460         SapModel.RespCombo.Add comboName, comboType
461         SapModel.RespCombo.SetCaseList comboName, comboLoadType, "Dead",
462     1.2
463         SapModel.RespCombo.SetCaseList comboName, comboLoadType, "SIDL",
464     1.2
465         SapModel.RespCombo.SetCaseList comboName, comboLoadType, "Live", 1
466         SapModel.RespCombo.SetCaseList comboName, comboLoadType, "Live
467     Roof", 0.5
468         SapModel.RespCombo.SetCaseList comboName, comboLoadType, "Wind", 1
469
470         comboName = "LRFD 4-2"
471         SapModel.RespCombo.Add comboName, comboType
472         SapModel.RespCombo.SetCaseList comboName, comboLoadType, "Dead",
473     1.2
474         SapModel.RespCombo.SetCaseList comboName, comboLoadType, "SIDL",
475     1.2
476         SapModel.RespCombo.SetCaseList comboName, comboLoadType, "Live", 1
477         SapModel.RespCombo.SetCaseList comboName, comboLoadType, "Live
478     Roof", 0.5
479         SapModel.RespCombo.SetCaseList comboName, comboLoadType, "Wind", 1
480
481         comboName = "LRFD 4-3"
482         SapModel.RespCombo.Add comboName, comboType
483         SapModel.RespCombo.SetCaseList comboName, comboLoadType, "Dead",
484     1.2
485         SapModel.RespCombo.SetCaseList comboName, comboLoadType, "SIDL",
486     1.2
487         SapModel.RespCombo.SetCaseList comboName, comboLoadType, "Live", 1
488         SapModel.RespCombo.SetCaseList comboName, comboLoadType, "Live
489     Roof", 0.5
490         SapModel.RespCombo.SetCaseList comboName, comboLoadType, "Wind",
491     0.75
```

```
492
493         comboName = "LRFD 4-4"
494         SapModel.RespCombo.Add comboName, comboType
495         SapModel.RespCombo.SetCaseList comboName, comboLoadType, "Dead",
496     1.2
497         SapModel.RespCombo.SetCaseList comboName, comboLoadType, "SIDL",
498     1.2
499         SapModel.RespCombo.SetCaseList comboName, comboLoadType, "Live", 1
500         SapModel.RespCombo.SetCaseList comboName, comboLoadType, "Live
501     Roof", 0.5
502         SapModel.RespCombo.SetCaseList comboName, comboLoadType, "Wind", 1
503
504         comboName = "LRFD 4-5"
505         SapModel.RespCombo.Add comboName, comboType
506         SapModel.RespCombo.SetCaseList comboName, comboLoadType, "Dead",
507     1.2
508         SapModel.RespCombo.SetCaseList comboName, comboLoadType, "SIDL",
509     1.2
510         SapModel.RespCombo.SetCaseList comboName, comboLoadType, "Live", 1
511         SapModel.RespCombo.SetCaseList comboName, comboLoadType, "Live
512     Roof", 0.5
513         SapModel.RespCombo.SetCaseList comboName, comboLoadType, "Wind", 1
514
515         comboName = "LRFD 4-6"
516         SapModel.RespCombo.Add comboName, comboType
517         SapModel.RespCombo.SetCaseList comboName, comboLoadType, "Dead",
518     1.2
519         SapModel.RespCombo.SetCaseList comboName, comboLoadType, "SIDL",
520     1.2
521         SapModel.RespCombo.SetCaseList comboName, comboLoadType, "Live", 1
522         SapModel.RespCombo.SetCaseList comboName, comboLoadType, "Live
523     Roof", 0.5
524         SapModel.RespCombo.SetCaseList comboName, comboLoadType, "Wind",
525     0.75
526
527         comboName = "LRFD 5-1"
528         SapModel.RespCombo.Add comboName, comboType
529         SapModel.RespCombo.SetCaseList comboName, comboLoadType, "Dead",
530     0.9
531         SapModel.RespCombo.SetCaseList comboName, comboLoadType, "SIDL",
532     0.9
533         SapModel.RespCombo.SetCaseList comboName, comboLoadType, "Wind", 1
534
535         comboName = "LRFD 5-2"
536         SapModel.RespCombo.Add comboName, comboType
537         SapModel.RespCombo.SetCaseList comboName, comboLoadType, "Dead",
538     0.9
539         SapModel.RespCombo.SetCaseList comboName, comboLoadType, "SIDL",
540     0.9
541         SapModel.RespCombo.SetCaseList comboName, comboLoadType, "Wind", 1
542
543         comboName = "LRFD 5-3"
544         SapModel.RespCombo.Add comboName, comboType
545         SapModel.RespCombo.SetCaseList comboName, comboLoadType, "Dead",
546     0.9
547         SapModel.RespCombo.SetCaseList comboName, comboLoadType, "SIDL",
548     0.9
```

```
    SapModel.RespCombo.SetCaseList comboName, comboLoadType, "Wind",
0.75

    Dim MyLoadPat() As String, MySF() As Double
    ReDim MyLoadPat(2)
    ReDim MySF(2)
    MyLoadPat(0) = "Dead"
    MyLoadPat(1) = "Live"
    MyLoadPat(2) = "SIDL"
    MySF(0) = 1#
    MySF(1) = 0.3
    MySF(2) = 1#
    ret = SapModel.PropMaterial.SetMassSource(3, 3, MyLoadPat, MySF)
    ret = SapModel.View.RefreshView(0, False)
    Application.StatusBar = False
    Application.ScreenUpdating = True
    Application.Calculation = xlCalculationAutomatic
    MsgBox "Model Sudah Jadi nih Bos! ", vbInformation, "EXCEL ke
ETABS"
    Exit Sub

ErrHandler:
    Application.StatusBar = False
    Application.ScreenUpdating = True
    Application.Calculation = xlCalculationAutomatic
    MsgBox "Error: " & Err.Description, vbCritical, "ETABS Complete
Model"
End Sub
```

Lampiran 2 Module 2 (MASUKKAN_PEMBEBANAN)

```
Option Explicit

'=== Helpers ===
Private Function GetSapModel() As Object
    Dim h As Object, et As Object
    On Error Resume Next
    Set h = CreateObject("ETABSv1.Helper")
    If Not h Is Nothing Then Set et =
h.GetObject("CSI.ETABS.API.ETABSObject")
    On Error GoTo 0
    If Not et Is Nothing Then Set GetSapModel = et.SapModel
End Function

Private Function FloorQ(Optional ByVal def As Double = 2#) As Double
    Dim v As Variant: v = ThisWorkbook.Sheets("DATA
INPUT").Range("G33").Value
    If IsNumeric(v) And v > 0 Then FloorQ = v Else FloorQ = def
End Function

Private Function RoofQ(Optional ByVal def As Double = 0#) As Double
    Dim v As Variant: v = ThisWorkbook.Sheets("DATA
INPUT").Range("G34").Value
    If IsNumeric(v) And v > 0 Then RoofQ = v Else RoofQ = IIf(def > 0,
def, FloorQ(2#))
End Function

Private Function KeramikQ(Optional ByVal def As Double = 0#) As Double
```

```
    Dim v As Variant: v = ThisWorkbook.Sheets("DATA
INPUT").Range("G36").Value
    If IsNumeric(v) And v >= 0 Then KeramikQ = v Else KeramikQ = def
End Function

Private Function PlafonQ(Optional ByVal def As Double = 0#) As Double
    Dim v As Variant: v = ThisWorkbook.Sheets("DATA
INPUT").Range("G37").Value
    If IsNumeric(v) And v >= 0 Then PlafonQ = v Else PlafonQ = def
End Function

Private Function JumlahSIDLQ(Optional ByVal def As Double = 0#) As
Double
    Dim v As Variant: v = ThisWorkbook.Sheets("DATA
INPUT").Range("G38").Value
    If IsNumeric(v) And v >= 0 Then JumlahSIDLQ = v Else JumlahSIDLQ =
KeramikQ() + PlafonQ()
End Function

Private Function BeamWallQ(Optional ByVal def As Double = 0#) As
Double
    Dim v As Variant: v = ThisWorkbook.Sheets("DATA
INPUT").Range("G40").Value
    If IsNumeric(v) And v >= 0 Then BeamWallQ = v Else BeamWallQ = def
End Function

' === PEMBEBANAN PELAT DAN BALOK ===
Public Sub MASUKKAN_PEMBEBANAN()
    Dim m As Object, ws As Worksheet, ret As Long
    Dim qFloor As Double, qRoof As Double, qKeramik As Double, qPlafon
As Double, qJumlah As Double, qWall As Double
    Dim nStory As Long, nGX As Long, nGY As Long, h As Double
    Dim i As Long, j As Long, k As Long, id As Long, frameID As Long
    Dim okLive As Long, okSIDL As Long, totSlab As Long
    Dim okBeamX As Long, okBeamY As Long, totBeam As Long
    Dim frameLoad As Double

    Set m = GetSapModel()
    If m Is Nothing Then
        MsgBox "ETABS tidak ditemukan.", vbExclamation
        Exit Sub
    End If
    Set ws = ThisWorkbook.Sheets("DATA INPUT")

    ' Ambil dari data Input Excel
    qFloor = FloorQ(2#)
    qRoof = RoofQ(qFloor)
    qKeramik = KeramikQ(0#)
    qPlafon = PlafonQ(0#)
    qJumlah = JumlahSIDLQ(0#)
    qWall = BeamWallQ(0#)
    nStory = CLng(val(ws.Range("G12").Value))
    nGX = CLng(val(ws.Range("G8").Value))
    nGY = CLng(val(ws.Range("G9").Value))
    h = CDbl(val(ws.Range("G13").Value))

    ' === PEMBEBANAN PELAT ===
    id = 1
```

```
86      For i = 1 To nStory
87          Dim qLiveUse As Double, qSIDLUse As Double
88          qLiveUse = IIf(i = nStory, qRoof, qFloor)
89          If i = nStory Then qSIDLUse = qPlafon Else qSIDLUse = qJumlah
90
91          For k = 1 To nGY - 1
92              For j = 1 To nGX - 1
93                  ret = m.AreaObj.SetLoadUniform(CStr(id), "Live",
94  qLiveUse, 10, True)
95                  If ret = 0 Then okLive = okLive + 1
96                  If qSIDLUse > 0 Then
97                      ret = m.AreaObj.SetLoadUniform(CStr(id), "SIDL",
98  qSIDLUse, 10, True)
99                      If ret = 0 Then okSIDL = okSIDL + 1
100                 End If
101                 totSlab = totSlab + 1
102                 id = id + 1
103             Next j
104         Next k
105     Next i
106
107     ' === PEMBEBANAN BALOK ===
108     If qWall > 0 Then
109         frameID = nStory * nGX * nGY + 1
110
111         For i = 0 To nStory - 1
112             frameLoad = IIf(i = nStory - 1, qWall * h / 2, qWall * h)
113
114             ' Balok arah X
115             For j = 0 To nGX - 2
116                 ret = m.FrameObj.SetLoadDistributed(CStr(frameID),
117  "SIDL", 1, 10, 0, 1, frameLoad, frameLoad)
118                 If ret = 0 Then okBeamX = okBeamX + 1
119                 totBeam = totBeam + 1
120                 frameID = frameID + 1
121             Next j
122             frameID = frameID + (nGX - 1) * (nGY - 2)
123             For j = 0 To nGX - 2
124                 ret = m.FrameObj.SetLoadDistributed(CStr(frameID),
125  "SIDL", 1, 10, 0, 1, frameLoad, frameLoad)
126                 If ret = 0 Then okBeamX = okBeamX + 1
127                 totBeam = totBeam + 1
128                 frameID = frameID + 1
129             Next j
130         Next i
131
132         ' Balok arah Y
133         For i = 0 To nStory - 1
134             frameLoad = IIf(i = nStory - 1, qWall * h / 2, qWall * h)
135             For j = 0 To nGY - 2
136                 ret = m.FrameObj.SetLoadDistributed(CStr(frameID),
137  "SIDL", 1, 10, 0, 1, frameLoad, frameLoad)
138                 If ret = 0 Then okBeamY = okBeamY + 1
139                 totBeam = totBeam + 1
140                 frameID = frameID + (nGX - 1)
141                 ret = m.FrameObj.SetLoadDistributed(CStr(frameID),
142  "SIDL", 1, 10, 0, 1, frameLoad, frameLoad)
143                 If ret = 0 Then okBeamY = okBeamY + 1
```

```
                    totBeam = totBeam + 1
                    frameID = frameID + 1
                Next j
            Next i
        End If

        m.View.RefreshView 0, False
        Dim msg As String
        msg = "=== PEMBEBANAN SUDAH DIMASUKKAN ===" & vbCrLf & vbCrLf
        msg = msg & "PELAT - LIVE LOADS:" & vbCrLf
        msg = msg & "• Lantai (G33): " & Format(qFloor, "0.00") & " kN/m²"
& vbCrLf
        msg = msg & "• Atap (G34): " & Format(qRoof, "0.00") & " kN/m²" &
vbCrLf
        msg = msg & "• Pelat OK: " & okLive & "/" & totSlab & vbCrLf &
vbCrLf
        msg = msg & "PELAT - SIDL LOADS:" & vbCrLf
        msg = msg & "• Keramik (G36): " & Format(qKeramik, "0.00") & "
kN/m²" & vbCrLf
        msg = msg & "• Plafon (G37): " & Format(qPlafon, "0.00") & "
kN/m²" & vbCrLf
        msg = msg & "• Jumlah (G38): " & Format(qJumlah, "0.00") & "
kN/m²" & vbCrLf
        msg = msg & "• Pelat OK: " & okSIDL & "/" & totSlab & vbCrLf &
vbCrLf
        If qWall > 0 Then
            msg = msg & "BALOK - BEBAN DINDING (SIDL):" & vbCrLf
            msg = msg & "• Beban (G40): " & Format(qWall, "0.00") & "
kN/m/m" & vbCrLf
            msg = msg & "• Lt.1-" & (nStory - 1) & ": " & Format(qWall *
h, "0.00") & " kN/m" & vbCrLf
            msg = msg & "• Lt." & nStory & " (Atap): " & Format(qWall * h
/ 2, "0.00") & " kN/m" & vbCrLf
            msg = msg & "• Balok OK: " & (okBeamX + okBeamY) & "/" &
totBeam
        Else
            msg = msg & "BALOK: Tidak ada beban (G40 = 0)"
        End If

        MsgBox msg, vbInformation, "Complete Loading Applied"
End Sub

```

Lampiran 3 Module 3 (APPLY_RESPONSE_SPECTRUM)

```
Option Explicit

Private Function GetSapModel() As Object
    Dim h As Object, et As Object
    On Error Resume Next
    Set h = CreateObject("ETABSv1.Helper")
    If Not h Is Nothing Then Set et =
h.GetObject("CSI.ETABS.API.ETABSObject")
    On Error GoTo 0
    If Not et Is Nothing Then Set GetSapModel = et.SapModel
End Function

Public Sub APPLY_RESPONSE_SPECTRUM()
    Dim m As Object, ws As Worksheet, ret As Long

```

```
    Set m = GetSapModel()
    If m Is Nothing Then
        MsgBox "ETABS tidak ditemukan.", vbExclamation
        Exit Sub
    End If
    Set ws = ThisWorkbook.Sheets("DATA INPUT")

    Dim soilType As String, siteClass As Long
    Dim pga As Double, ss As Double, s1 As Double, tL As Double, ts As
Double
    Dim fa As Double, fv As Double

    soilType = Trim(ws.Range("G47").Value)
    pga = CDbl(ws.Range("G48").Value)
    ss = CDbl(ws.Range("G49").Value)
    s1 = CDbl(ws.Range("G50").Value)
    tL = CDbl(ws.Range("G51").Value)
    ts = CDbl(ws.Range("G53").Value)

    Select Case UCase(soilType)
        Case "A": siteClass = 1
        Case "B": siteClass = 2
        Case "C": siteClass = 3
        Case "D": siteClass = 4
        Case "E": siteClass = 5
        Case Else: siteClass = 4
    End Select

    Dim ParamsOption As Integer
    Dim Latitude As Double, Longitude As Double
    Dim Island As Integer, LimitState As Integer, UsageClass As
Integer
    Dim NomLife As Double, PeakAccel As Double, F0 As Double, Tcs As
Double
    Dim SpecType As Integer, SoilType2 As Integer, Topography As
Integer
    Dim hRatio As Double, Damping As Double, q As Double

    ParamsOption = 3
    Latitude = 0
    Longitude = 0
    Island = 1
    LimitState = 3
    UsageClass = 2
    NomLife = 50
    PeakAccel = pga
    F0 = IIf(pga > 0, ss / pga, 2.5)
    Tcs = ts
    SpecType = 3
    SoilType2 = siteClass
    Topography = 1
    hRatio = 1
    Damping = 5
    q = 1

    ret = m.Func.FuncRS.SetNTC2018("Response Spectrum", ParamsOption,
Latitude, Longitude, Island, LimitState, UsageClass, NomLife,
```

```
PeakAccel, F0, Tcs, SpecType, SoilType2, Topography, hRatio, Damping,
q)
    Dim nStory As Long, storyH As Double
    nStory = CLng(val(ws.Range("G12").Value))
    storyH = CDbl(val(ws.Range("G13").Value))

    Dim topZ As Double, bottomZ As Double
    topZ = nStory * storyH
    bottomZ = 0

    Dim nDir() As Boolean
    ReDim nDir(6)

    Dim eccen As Double, periodFlag As Long, ctType As Long
    Dim userT As Double, userZ As Boolean
    Dim r As Double, omega As Double, cd As Double, impFactor As
Double

    eccen = 0.05
    periodFlag = 2
    ctType = 1
    userT = 0
    userZ = False
    r = 8#
    omega = 3#
    cd = 5.5
    impFactor = 1#

    ret = m.LoadPatterns.Add("Spec X", 5, 0)
    nDir(1) = True
    nDir(2) = False
    nDir(3) = True
    nDir(4) = False
    nDir(5) = True
    nDir(6) = False
    ret = m.LoadPatterns.AutoSeismic.SetASCE716("Spec X", nDir, eccen,
periodFlag, ctType, userT, userZ, topZ, bottomZ, r, omega, cd,
impFactor, ss, s1, tL, siteClass, fa, fv)

    ret = m.LoadPatterns.Add("Spec Y", 5, 0)
    nDir(1) = False
    nDir(2) = True
    nDir(3) = False
    nDir(4) = True
    nDir(5) = False
    nDir(6) = True
    ret = m.LoadPatterns.AutoSeismic.SetASCE716("Spec Y", nDir, eccen,
periodFlag, ctType, userT, userZ, topZ, bottomZ, r, omega, cd,
impFactor, ss, s1, tL, siteClass, fa, fv)

    ret = m.LoadCases.ResponseSpectrum.SetCase("RSx")
    ret = m.LoadCases.ResponseSpectrum.SetCase("RSy")

    Dim SDS As Double
    SDS = ws.Range("G54").Value2

    Dim rho As Double
    rho = 1.3
```

```
131
132         Dim Ohm As Double
133         Ohm = CDbl(val(ws.Range("G57").Value))
134
135         Dim comboName As String, comboType As Long
136         comboType = 0
137
138         comboName = "LRFD 6-1"
139         ret = m.RespCombo.Add(comboName, comboType)
140         ret = m.RespCombo.SetCaseList(comboName, 0, "Dead", 1.2 + 0.2 *
141     SDS)
142         ret = m.RespCombo.SetCaseList(comboName, 0, "SIDL", 1.2 + 0.2 *
143     SDS)
144         ret = m.RespCombo.SetCaseList(comboName, 0, "Live", 1#)
145         ret = m.RespCombo.SetCaseList(comboName, 0, "Spec X", rho)
146         ret = m.RespCombo.SetCaseList(comboName, 0, "Spec Y", 0.3 * rho)
147
148         comboName = "LRFD 6-2"
149         ret = m.RespCombo.Add(comboName, comboType)
150         ret = m.RespCombo.SetCaseList(comboName, 0, "Dead", 1.2 + 0.2 *
151     SDS)
152         ret = m.RespCombo.SetCaseList(comboName, 0, "SIDL", 1.2 + 0.2 *
153     SDS)
154         ret = m.RespCombo.SetCaseList(comboName, 0, "Live", 1#)
155         ret = m.RespCombo.SetCaseList(comboName, 0, "Spec X", rho)
156         ret = m.RespCombo.SetCaseList(comboName, 0, "Spec Y", -1 * 0.3 *
157     rho)
158
159         comboName = "LRFD 6-3"
160         ret = m.RespCombo.Add(comboName, comboType)
161         ret = m.RespCombo.SetCaseList(comboName, 0, "Dead", 1.2 + 0.2 *
162     SDS)
163         ret = m.RespCombo.SetCaseList(comboName, 0, "SIDL", 1.2 + 0.2 *
164     SDS)
165         ret = m.RespCombo.SetCaseList(comboName, 0, "Live", 1#)
166         ret = m.RespCombo.SetCaseList(comboName, 0, "Spec X", -1 * rho)
167         ret = m.RespCombo.SetCaseList(comboName, 0, "Spec Y", 0.3 * rho)
168
169         comboName = "LRFD 6-4"
170         ret = m.RespCombo.Add(comboName, comboType)
171         ret = m.RespCombo.SetCaseList(comboName, 0, "Dead", 1.2 + 0.2 *
172     SDS)
173         ret = m.RespCombo.SetCaseList(comboName, 0, "SIDL", 1.2 + 0.2 *
174     SDS)
175         ret = m.RespCombo.SetCaseList(comboName, 0, "Live", 1#)
176         ret = m.RespCombo.SetCaseList(comboName, 0, "Spec X", -1 * rho)
177         ret = m.RespCombo.SetCaseList(comboName, 0, "Spec Y", -1 * 0.3 *
178     rho)
179
180         comboName = "LRFD 6-5"
181         ret = m.RespCombo.Add(comboName, comboType)
182         ret = m.RespCombo.SetCaseList(comboName, 0, "Dead", 1.2 + 0.2 *
183     SDS)
184         ret = m.RespCombo.SetCaseList(comboName, 0, "SIDL", 1.2 + 0.2 *
185     SDS)
186         ret = m.RespCombo.SetCaseList(comboName, 0, "Live", 1#)
187         ret = m.RespCombo.SetCaseList(comboName, 0, "Spec X", 0.3 * rho)
188         ret = m.RespCombo.SetCaseList(comboName, 0, "Spec Y", rho)
```

```
189
190         comboName = "LRFD 6-6"
191         ret = m.RespCombo.Add(comboName, comboType)
192         ret = m.RespCombo.SetCaseList(comboName, 0, "Dead", 1.2 + 0.2 *
193     SDS)
194         ret = m.RespCombo.SetCaseList(comboName, 0, "SIDL", 1.2 + 0.2 *
195     SDS)
196         ret = m.RespCombo.SetCaseList(comboName, 0, "Live", 1#)
197         ret = m.RespCombo.SetCaseList(comboName, 0, "Spec X", -1 * 0.3 *
198     rho)
199         ret = m.RespCombo.SetCaseList(comboName, 0, "Spec Y", rho)
200
201         comboName = "LRFD 6-7"
202         ret = m.RespCombo.Add(comboName, comboType)
203         ret = m.RespCombo.SetCaseList(comboName, 0, "Dead", 1.2 + 0.2 *
204     SDS)
205         ret = m.RespCombo.SetCaseList(comboName, 0, "SIDL", 1.2 + 0.2 *
206     SDS)
207         ret = m.RespCombo.SetCaseList(comboName, 0, "Live", 1#)
208         ret = m.RespCombo.SetCaseList(comboName, 0, "Spec X", 0.3 * rho)
209         ret = m.RespCombo.SetCaseList(comboName, 0, "Spec Y", -1 * rho)
210
211         comboName = "LRFD 6-8"
212         ret = m.RespCombo.Add(comboName, comboType)
213         ret = m.RespCombo.SetCaseList(comboName, 0, "Dead", 1.2 + 0.2 *
214     SDS)
215         ret = m.RespCombo.SetCaseList(comboName, 0, "SIDL", 1.2 + 0.2 *
216     SDS)
217         ret = m.RespCombo.SetCaseList(comboName, 0, "Live", 1#)
218         ret = m.RespCombo.SetCaseList(comboName, 0, "Spec X", -1 * 0.3 *
219     rho)
220         ret = m.RespCombo.SetCaseList(comboName, 0, "Spec Y", -1 * rho)
221
222         comboName = "LRFD 7-1"
223         ret = m.RespCombo.Add(comboName, comboType)
224         ret = m.RespCombo.SetCaseList(comboName, 0, "Dead", 0.9 - 0.2 *
225     SDS)
226         ret = m.RespCombo.SetCaseList(comboName, 0, "SIDL", 0.9 - 0.2 *
227     SDS)
228         ret = m.RespCombo.SetCaseList(comboName, 0, "Spec X", rho)
229         ret = m.RespCombo.SetCaseList(comboName, 0, "Spec Y", 0.3 * rho)
230         comboName = "LRFD 7-2"
231         ret = m.RespCombo.Add(comboName, comboType)
232         ret = m.RespCombo.SetCaseList(comboName, 0, "Dead", 0.9 - 0.2 *
233     SDS)
234         ret = m.RespCombo.SetCaseList(comboName, 0, "SIDL", 0.9 - 0.2 *
235     SDS)
236         ret = m.RespCombo.SetCaseList(comboName, 0, "Spec X", rho)
237         ret = m.RespCombo.SetCaseList(comboName, 0, "Spec Y", -1 * 0.3 *
238     rho)
239
240         comboName = "LRFD 7-3"
241         ret = m.RespCombo.Add(comboName, comboType)
242         ret = m.RespCombo.SetCaseList(comboName, 0, "Dead", 0.9 - 0.2 *
243     SDS)
244         ret = m.RespCombo.SetCaseList(comboName, 0, "SIDL", 0.9 - 0.2 *
245     SDS)
246         ret = m.RespCombo.SetCaseList(comboName, 0, "Spec X", -1 * rho)
```

```
247         ret = m.RespCombo.SetCaseList(comboName, 0, "Spec Y", 0.3 * rho)
248
249         comboName = "LRFD 7-4"
250         ret = m.RespCombo.Add(comboName, comboType)
251         ret = m.RespCombo.SetCaseList(comboName, 0, "Dead", 0.9 - 0.2 *
252     SDS)
253         ret = m.RespCombo.SetCaseList(comboName, 0, "SIDL", 0.9 - 0.2 *
254     SDS)
255         ret = m.RespCombo.SetCaseList(comboName, 0, "Spec X", -1 * rho)
256         ret = m.RespCombo.SetCaseList(comboName, 0, "Spec Y", -1 * 0.3 *
257     rho)
258
259         comboName = "LRFD 7-5"
260         ret = m.RespCombo.Add(comboName, comboType)
261         ret = m.RespCombo.SetCaseList(comboName, 0, "Dead", 0.9 - 0.2 *
262     SDS)
263         ret = m.RespCombo.SetCaseList(comboName, 0, "SIDL", 0.9 - 0.2 *
264     SDS)
265         ret = m.RespCombo.SetCaseList(comboName, 0, "Spec X", 0.3 * rho)
266         ret = m.RespCombo.SetCaseList(comboName, 0, "Spec Y", rho)
267
268         comboName = "LRFD 7-6"
269         ret = m.RespCombo.Add(comboName, comboType)
270         ret = m.RespCombo.SetCaseList(comboName, 0, "Dead", 0.9 - 0.2 *
271     SDS)
272         ret = m.RespCombo.SetCaseList(comboName, 0, "SIDL", 0.9 - 0.2 *
273     SDS)
274         ret = m.RespCombo.SetCaseList(comboName, 0, "Spec X", -1 * 0.3 *
275     rho)
276         ret = m.RespCombo.SetCaseList(comboName, 0, "Spec Y", rho)
277
278         comboName = "LRFD 7-7"
279         ret = m.RespCombo.Add(comboName, comboType)
280         ret = m.RespCombo.SetCaseList(comboName, 0, "Dead", 0.9 - 0.2 *
281     SDS)
282         ret = m.RespCombo.SetCaseList(comboName, 0, "SIDL", 0.9 - 0.2 *
283     SDS)
284         ret = m.RespCombo.SetCaseList(comboName, 0, "Spec X", 0.3 * rho)
285         ret = m.RespCombo.SetCaseList(comboName, 0, "Spec Y", -1 * rho)
286
287         comboName = "LRFD 7-8"
288         ret = m.RespCombo.Add(comboName, comboType)
289         ret = m.RespCombo.SetCaseList(comboName, 0, "Dead", 0.9 - 0.2 *
290     SDS)
291         ret = m.RespCombo.SetCaseList(comboName, 0, "SIDL", 0.9 - 0.2 *
292     SDS)
293         ret = m.RespCombo.SetCaseList(comboName, 0, "Spec X", -1 * 0.3 *
294     rho)
295         ret = m.RespCombo.SetCaseList(comboName, 0, "Spec Y", -1 * rho)
296
297         comboName = "Service"
298         ret = m.RespCombo.Add(comboName, comboType)
299         ret = m.RespCombo.SetCaseList(comboName, 0, "Dead", 1#)
300         ret = m.RespCombo.SetCaseList(comboName, 0, "SIDL", 1#)
301         ret = m.RespCombo.SetCaseList(comboName, 0, "Live", 1#)
302
303         comboName = "ASD G1"
304         ret = m.RespCombo.Add(comboName, comboType)
```

```
305        ret = m.RespCombo.SetCaseList(comboName, 0, "Dead", 1#)
306        ret = m.RespCombo.SetCaseList(comboName, 0, "SIDL", 1#)
307
308        comboName = "ASD G2"
309        ret = m.RespCombo.Add(comboName, comboType)
310        ret = m.RespCombo.SetCaseList(comboName, 0, "Dead", 1#)
311        ret = m.RespCombo.SetCaseList(comboName, 0, "SIDL", 1#)
312        ret = m.RespCombo.SetCaseList(comboName, 0, "Live", 1#)
313
314        comboName = "ASD G3"
315        ret = m.RespCombo.Add(comboName, comboType)
316        ret = m.RespCombo.SetCaseList(comboName, 0, "Dead", 1#)
317        ret = m.RespCombo.SetCaseList(comboName, 0, "SIDL", 1#)
318        ret = m.RespCombo.SetCaseList(comboName, 0, "Live Roof", 1#)
319
320        comboName = "ASD G4"
321        ret = m.RespCombo.Add(comboName, comboType)
322        ret = m.RespCombo.SetCaseList(comboName, 0, "Dead", 1#)
323        ret = m.RespCombo.SetCaseList(comboName, 0, "SIDL", 1#)
324        ret = m.RespCombo.SetCaseList(comboName, 0, "Live Roof", 1#)
325
326        comboName = "ASD G5"
327        ret = m.RespCombo.Add(comboName, comboType)
328        ret = m.RespCombo.SetCaseList(comboName, 0, "Dead", 1#)
329        ret = m.RespCombo.SetCaseList(comboName, 0, "SIDL", 1#)
330        ret = m.RespCombo.SetCaseList(comboName, 0, "Live", 0.75)
331        ret = m.RespCombo.SetCaseList(comboName, 0, "Live Roof", 0.75)
332
333        comboName = "ASD G6"
334        ret = m.RespCombo.Add(comboName, comboType)
335        ret = m.RespCombo.SetCaseList(comboName, 0, "Dead", 1#)
336        ret = m.RespCombo.SetCaseList(comboName, 0, "SIDL", 1#)
337        ret = m.RespCombo.SetCaseList(comboName, 0, "Live", 0.75)
338        ret = m.RespCombo.SetCaseList(comboName, 0, "Live Roof", 0.75)
339
340        comboName = "ASD G7"
341        ret = m.RespCombo.Add(comboName, comboType)
342        ret = m.RespCombo.SetCaseList(comboName, 0, "Dead", 1#)
343        ret = m.RespCombo.SetCaseList(comboName, 0, "SIDL", 1#)
344        ret = m.RespCombo.SetCaseList(comboName, 0, "Wind", 0.6)
345
346        comboName = "ASD G8"
347        ret = m.RespCombo.Add(comboName, comboType)
348        ret = m.RespCombo.SetCaseList(comboName, 0, "Dead", 1#)
349        ret = m.RespCombo.SetCaseList(comboName, 0, "SIDL", 1#)
350        ret = m.RespCombo.SetCaseList(comboName, 0, "Wind", 0.6)
351
352        comboName = "ASD G9"
353        ret = m.RespCombo.Add(comboName, comboType)
354        ret = m.RespCombo.SetCaseList(comboName, 0, "Dead", 1#)
355        ret = m.RespCombo.SetCaseList(comboName, 0, "SIDL", 1#)
356        ret = m.RespCombo.SetCaseList(comboName, 0, "Wind", 0.45)
357
358        comboName = "ASD G10"
359        ret = m.RespCombo.Add(comboName, comboType)
360        ret = m.RespCombo.SetCaseList(comboName, 0, "Dead", 1#)
361        ret = m.RespCombo.SetCaseList(comboName, 0, "SIDL", 1#)
362        ret = m.RespCombo.SetCaseList(comboName, 0, "Live", 0.75)
```

```
363        ret = m.RespCombo.SetCaseList(comboName, 0, "Live Roof", 0.75)
364        ret = m.RespCombo.SetCaseList(comboName, 0, "Wind", 0.45)
365
366        comboName = "ASD G11"
367        ret = m.RespCombo.Add(comboName, comboType)
368        ret = m.RespCombo.SetCaseList(comboName, 0, "Dead", 1#)
369        ret = m.RespCombo.SetCaseList(comboName, 0, "SIDL", 1#)
370        ret = m.RespCombo.SetCaseList(comboName, 0, "Live", 0.75)
371        ret = m.RespCombo.SetCaseList(comboName, 0, "Live Roof", 0.75)
372        ret = m.RespCombo.SetCaseList(comboName, 0, "Wind", 0.45)
373
374        comboName = "ASD G12"
375        ret = m.RespCombo.Add(comboName, comboType)
376        ret = m.RespCombo.SetCaseList(comboName, 0, "Dead", 1#)
377        ret = m.RespCombo.SetCaseList(comboName, 0, "SIDL", 1#)
378        ret = m.RespCombo.SetCaseList(comboName, 0, "Live", 0.75)
379        ret = m.RespCombo.SetCaseList(comboName, 0, "Live Roof", 0.75)
380        ret = m.RespCombo.SetCaseList(comboName, 0, "Wind", 0.3375)
381
382        comboName = "ASD G13"
383        ret = m.RespCombo.Add(comboName, comboType)
384        ret = m.RespCombo.SetCaseList(comboName, 0, "Dead", 1#)
385        ret = m.RespCombo.SetCaseList(comboName, 0, "SIDL", 1#)
386        ret = m.RespCombo.SetCaseList(comboName, 0, "Live", 0.75)
387        ret = m.RespCombo.SetCaseList(comboName, 0, "Live Roof", 0.75)
388        ret = m.RespCombo.SetCaseList(comboName, 0, "Wind", 0.45)
389
390        comboName = "ASD G14"
391        ret = m.RespCombo.Add(comboName, comboType)
392        ret = m.RespCombo.SetCaseList(comboName, 0, "Dead", 1#)
393        ret = m.RespCombo.SetCaseList(comboName, 0, "SIDL", 1#)
394        ret = m.RespCombo.SetCaseList(comboName, 0, "Live", 0.75)
395        ret = m.RespCombo.SetCaseList(comboName, 0, "Live Roof", 0.75)
396        ret = m.RespCombo.SetCaseList(comboName, 0, "Wind", 0.45)
397
398        comboName = "ASD G15"
399        ret = m.RespCombo.Add(comboName, comboType)
400        ret = m.RespCombo.SetCaseList(comboName, 0, "Dead", 1#)
401        ret = m.RespCombo.SetCaseList(comboName, 0, "SIDL", 1#)
402        ret = m.RespCombo.SetCaseList(comboName, 0, "Live", 0.75)
403        ret = m.RespCombo.SetCaseList(comboName, 0, "Live Roof", 0.75)
404        ret = m.RespCombo.SetCaseList(comboName, 0, "Wind", 0.3375)
405
406        comboName = "ASD G16"
407        ret = m.RespCombo.Add(comboName, comboType)
408        ret = m.RespCombo.SetCaseList(comboName, 0, "Dead", 0.6)
409        ret = m.RespCombo.SetCaseList(comboName, 0, "SIDL", 0.6)
410        ret = m.RespCombo.SetCaseList(comboName, 0, "Wind", 0.6)
411
412        comboName = "ASD G17"
413        ret = m.RespCombo.Add(comboName, comboType)
414        ret = m.RespCombo.SetCaseList(comboName, 0, "Dead", 0.6)
415        ret = m.RespCombo.SetCaseList(comboName, 0, "SIDL", 0.6)
416        ret = m.RespCombo.SetCaseList(comboName, 0, "Wind", 0.6)
417
418        comboName = "ASD G18"
419        ret = m.RespCombo.Add(comboName, comboType)
420        ret = m.RespCombo.SetCaseList(comboName, 0, "Dead", 0.6)
```

```
    ret = m.RespCombo.SetCaseList(comboName, 0, "SIDL", 0.6)
    ret = m.RespCombo.SetCaseList(comboName, 0, "Wind", 0.45)

    comboName = "ASD N1"
    ret = m.RespCombo.Add(comboName, comboType)
    ret = m.RespCombo.SetCaseList(comboName, 0, "Dead", 1 + 0.7 * 0.2
* SDS)
    ret = m.RespCombo.SetCaseList(comboName, 0, "SIDL", 1 + 0.7 * 0.2
* SDS)
    ret = m.RespCombo.SetCaseList(comboName, 0, "Spec X", 0.7 * 0.3 *
rho)
    ret = m.RespCombo.SetCaseList(comboName, 0, "Spec Y", 0.7 * rho)

    comboName = "ASD N2"
    ret = m.RespCombo.Add(comboName, comboType)
    ret = m.RespCombo.SetCaseList(comboName, 0, "Dead", 1 + 0.7 * 0.2
* SDS)
    ret = m.RespCombo.SetCaseList(comboName, 0, "SIDL", 1 + 0.7 * 0.2
* SDS)
    ret = m.RespCombo.SetCaseList(comboName, 0, "Spec X", 0.7 * 0.3 *
rho)
    ret = m.RespCombo.SetCaseList(comboName, 0, "Spec Y", -1 * 0.7 *
rho)

    comboName = "ASD N3"
    ret = m.RespCombo.Add(comboName, comboType)
    ret = m.RespCombo.SetCaseList(comboName, 0, "Dead", 1 + 0.7 * 0.2
* SDS)
    ret = m.RespCombo.SetCaseList(comboName, 0, "SIDL", 1 + 0.7 * 0.2
* SDS)
    ret = m.RespCombo.SetCaseList(comboName, 0, "Spec X", -1 * 0.7 *
0.3 * rho)
    ret = m.RespCombo.SetCaseList(comboName, 0, "Spec Y", 0.7 * rho)

    comboName = "ASD N4"
    ret = m.RespCombo.Add(comboName, comboType)
    ret = m.RespCombo.SetCaseList(comboName, 0, "Dead", 1 + 0.7 * 0.2
* SDS)
    ret = m.RespCombo.SetCaseList(comboName, 0, "SIDL", 1 + 0.7 * 0.2
* SDS)
    ret = m.RespCombo.SetCaseList(comboName, 0, "Spec X", -1 * 0.7 *
0.3 * rho)
    ret = m.RespCombo.SetCaseList(comboName, 0, "Spec Y", -1 * 0.7 *
rho)

    comboName = "ASD N5"
    ret = m.RespCombo.Add(comboName, comboType)
    ret = m.RespCombo.SetCaseList(comboName, 0, "Dead", 1 + 0.7 * 0.2
* SDS)
    ret = m.RespCombo.SetCaseList(comboName, 0, "SIDL", 1 + 0.7 * 0.2
* SDS)
    ret = m.RespCombo.SetCaseList(comboName, 0, "Spec X", 0.7 * rho)
    ret = m.RespCombo.SetCaseList(comboName, 0, "Spec Y", 0.7 * 0.3 *
rho)

    comboName = "ASD N6"
    ret = m.RespCombo.Add(comboName, comboType)
```

```
    ret = m.RespCombo.SetCaseList(comboName, 0, "Dead", 1 + 0.7 * 0.2
* SDS)
    ret = m.RespCombo.SetCaseList(comboName, 0, "SIDL", 1 + 0.7 * 0.2
* SDS)
    ret = m.RespCombo.SetCaseList(comboName, 0, "Spec X", -1 * 0.7 *
rho)
    ret = m.RespCombo.SetCaseList(comboName, 0, "Spec Y", 0.7 * 0.3 *
rho)

    comboName = "ASD N7"
    ret = m.RespCombo.Add(comboName, comboType)
    ret = m.RespCombo.SetCaseList(comboName, 0, "Dead", 1 + 0.7 * 0.2
* SDS)
    ret = m.RespCombo.SetCaseList(comboName, 0, "SIDL", 1 + 0.7 * 0.2
* SDS)
    ret = m.RespCombo.SetCaseList(comboName, 0, "Spec X", 0.7 * rho)
    ret = m.RespCombo.SetCaseList(comboName, 0, "Spec Y", -1 * 0.7 *
0.3 * rho)

    comboName = "ASD N8"
    ret = m.RespCombo.Add(comboName, comboType)
    ret = m.RespCombo.SetCaseList(comboName, 0, "Dead", 1 + 0.7 * 0.2
* SDS)
    ret = m.RespCombo.SetCaseList(comboName, 0, "SIDL", 1 + 0.7 * 0.2
* SDS)
    ret = m.RespCombo.SetCaseList(comboName, 0, "Spec X", -1 * 0.7 *
rho)
    ret = m.RespCombo.SetCaseList(comboName, 0, "Spec Y", -1 * 0.7 *
0.3 * rho)

    comboName = "ASD N9"
    ret = m.RespCombo.Add(comboName, comboType)
    ret = m.RespCombo.SetCaseList(comboName, 0, "Dead", 1 + 0.525 *
0.2 * SDS)
    ret = m.RespCombo.SetCaseList(comboName, 0, "SIDL", 1 + 0.525 *
0.2 * SDS)
    ret = m.RespCombo.SetCaseList(comboName, 0, "Live", 0.75)
    ret = m.RespCombo.SetCaseList(comboName, 0, "Spec X", 0.525 * 0.3
* rho)
    ret = m.RespCombo.SetCaseList(comboName, 0, "Spec Y", 0.525 * rho)

    comboName = "ASD N10"
    ret = m.RespCombo.Add(comboName, comboType)
    ret = m.RespCombo.SetCaseList(comboName, 0, "Dead", 1 + 0.525 *
0.2 * SDS)
    ret = m.RespCombo.SetCaseList(comboName, 0, "SIDL", 1 + 0.525 *
0.2 * SDS)
    ret = m.RespCombo.SetCaseList(comboName, 0, "Live", 0.75)
    ret = m.RespCombo.SetCaseList(comboName, 0, "Spec X", 0.525 * 0.3
* rho)
    ret = m.RespCombo.SetCaseList(comboName, 0, "Spec Y", -1 * 0.525 *
rho)

    comboName = "ASD N11"
    ret = m.RespCombo.Add(comboName, comboType)
    ret = m.RespCombo.SetCaseList(comboName, 0, "Dead", 1 + 0.525 *
0.2 * SDS)
```

```
    ret = m.RespCombo.SetCaseList(comboName, 0, "SIDL", 1 + 0.525 *
0.2 * SDS)
    ret = m.RespCombo.SetCaseList(comboName, 0, "Live", 0.75)
    ret = m.RespCombo.SetCaseList(comboName, 0, "Spec X", -1 * 0.525 *
0.3 * rho)
    ret = m.RespCombo.SetCaseList(comboName, 0, "Spec Y", 0.525 * rho)

    comboName = "ASD N12"
    ret = m.RespCombo.Add(comboName, comboType)
    ret = m.RespCombo.SetCaseList(comboName, 0, "Dead", 1 + 0.525 *
0.2 * SDS)
    ret = m.RespCombo.SetCaseList(comboName, 0, "SIDL", 1 + 0.525 *
0.2 * SDS)
    ret = m.RespCombo.SetCaseList(comboName, 0, "Live", 0.75)
    ret = m.RespCombo.SetCaseList(comboName, 0, "Spec X", -1 * 0.525 *
0.3 * rho)
    ret = m.RespCombo.SetCaseList(comboName, 0, "Spec Y", -1 * 0.525 *
rho)

    comboName = "ASD N13"
    ret = m.RespCombo.Add(comboName, comboType)
    ret = m.RespCombo.SetCaseList(comboName, 0, "Dead", 1 + 0.525 *
0.2 * SDS)
    ret = m.RespCombo.SetCaseList(comboName, 0, "SIDL", 1 + 0.525 *
0.2 * SDS)
    ret = m.RespCombo.SetCaseList(comboName, 0, "Live", 0.75)
    ret = m.RespCombo.SetCaseList(comboName, 0, "Spec X", 0.525 * rho)
    ret = m.RespCombo.SetCaseList(comboName, 0, "Spec Y", 0.525 * 0.3
* rho)

    comboName = "ASD N14"
    ret = m.RespCombo.Add(comboName, comboType)
    ret = m.RespCombo.SetCaseList(comboName, 0, "Dead", 1 + 0.525 *
0.2 * SDS)
    ret = m.RespCombo.SetCaseList(comboName, 0, "SIDL", 1 + 0.525 *
0.2 * SDS)
    ret = m.RespCombo.SetCaseList(comboName, 0, "Live", 0.75)
    ret = m.RespCombo.SetCaseList(comboName, 0, "Spec X", -1 * 0.525 *
rho)
    ret = m.RespCombo.SetCaseList(comboName, 0, "Spec Y", 0.525 * 0.3
* rho)

    comboName = "ASD N15"
    ret = m.RespCombo.Add(comboName, comboType)
    ret = m.RespCombo.SetCaseList(comboName, 0, "Dead", 1 + 0.525 *
0.2 * SDS)
    ret = m.RespCombo.SetCaseList(comboName, 0, "SIDL", 1 + 0.525 *
0.2 * SDS)
    ret = m.RespCombo.SetCaseList(comboName, 0, "Live", 0.75)
    ret = m.RespCombo.SetCaseList(comboName, 0, "Spec X", 0.525 * rho)
    ret = m.RespCombo.SetCaseList(comboName, 0, "Spec Y", -1 * 0.525 *
0.3 * rho)

    comboName = "ASD N16"
    ret = m.RespCombo.Add(comboName, comboType)
    ret = m.RespCombo.SetCaseList(comboName, 0, "Dead", 1 + 0.525 *
0.2 * SDS)
```

```
    ret = m.RespCombo.SetCaseList(comboName, 0, "SIDL", 1 + 0.525 *
0.2 * SDS)
    ret = m.RespCombo.SetCaseList(comboName, 0, "Live", 0.75)
    ret = m.RespCombo.SetCaseList(comboName, 0, "Spec X", -1 * 0.525 *
rho)
    ret = m.RespCombo.SetCaseList(comboName, 0, "Spec Y", -1 * 0.525 *
0.3 * rho)

    comboName = "ASD N17"
    ret = m.RespCombo.Add(comboName, comboType)
    ret = m.RespCombo.SetCaseList(comboName, 0, "Dead", 0.6 - 0.7 *
0.2 * SDS)
    ret = m.RespCombo.SetCaseList(comboName, 0, "SIDL", 0.6 - 0.7 *
0.2 * SDS)
    ret = m.RespCombo.SetCaseList(comboName, 0, "Spec X", 0.7 * 0.3 *
rho)
    ret = m.RespCombo.SetCaseList(comboName, 0, "Spec Y", 0.7 * rho)

    comboName = "ASD N18"
    ret = m.RespCombo.Add(comboName, comboType)
    ret = m.RespCombo.SetCaseList(comboName, 0, "Dead", 0.6 - 0.7 *
0.2 * SDS)
    ret = m.RespCombo.SetCaseList(comboName, 0, "SIDL", 0.6 - 0.7 *
0.2 * SDS)
    ret = m.RespCombo.SetCaseList(comboName, 0, "Spec X", 0.7 * 0.3 *
rho)
    ret = m.RespCombo.SetCaseList(comboName, 0, "Spec Y", -1 * 0.7 *
rho)

    comboName = "ASD N19"
    ret = m.RespCombo.Add(comboName, comboType)
    ret = m.RespCombo.SetCaseList(comboName, 0, "Dead", 0.6 - 0.7 *
0.2 * SDS)
    ret = m.RespCombo.SetCaseList(comboName, 0, "SIDL", 0.6 - 0.7 *
0.2 * SDS)
    ret = m.RespCombo.SetCaseList(comboName, 0, "Spec X", -1 * 0.7 *
0.3 * rho)
    ret = m.RespCombo.SetCaseList(comboName, 0, "Spec Y", 0.7 * rho)

    comboName = "ASD N20"
    ret = m.RespCombo.Add(comboName, comboType)
    ret = m.RespCombo.SetCaseList(comboName, 0, "Dead", 0.6 - 0.7 *
0.2 * SDS)
    ret = m.RespCombo.SetCaseList(comboName, 0, "SIDL", 0.6 - 0.7 *
0.2 * SDS)
    ret = m.RespCombo.SetCaseList(comboName, 0, "Spec X", -1 * 0.7 *
0.3 * rho)
    ret = m.RespCombo.SetCaseList(comboName, 0, "Spec Y", -1 * 0.7 *
rho)

    comboName = "ASD N21"
    ret = m.RespCombo.Add(comboName, comboType)
    ret = m.RespCombo.SetCaseList(comboName, 0, "Dead", 0.6 - 0.7 *
0.2 * SDS)
    ret = m.RespCombo.SetCaseList(comboName, 0, "SIDL", 0.6 - 0.7 *
0.2 * SDS)
    ret = m.RespCombo.SetCaseList(comboName, 0, "Spec X", 0.7 * rho)
```

```
    ret = m.RespCombo.SetCaseList(comboName, 0, "Spec Y", 0.7 * 0.3 *
rho)

    comboName = "ASD N22"
    ret = m.RespCombo.Add(comboName, comboType)
    ret = m.RespCombo.SetCaseList(comboName, 0, "Dead", 0.6 - 0.7 *
0.2 * SDS)
    ret = m.RespCombo.SetCaseList(comboName, 0, "SIDL", 0.6 - 0.7 *
0.2 * SDS)
    ret = m.RespCombo.SetCaseList(comboName, 0, "Spec X", -1 * 0.7 *
rho)
    ret = m.RespCombo.SetCaseList(comboName, 0, "Spec Y", 0.7 * 0.3 *
rho)

    comboName = "ASD N23"
    ret = m.RespCombo.Add(comboName, comboType)
    ret = m.RespCombo.SetCaseList(comboName, 0, "Dead", 0.6 - 0.7 *
0.2 * SDS)
    ret = m.RespCombo.SetCaseList(comboName, 0, "SIDL", 0.6 - 0.7 *
0.2 * SDS)
    ret = m.RespCombo.SetCaseList(comboName, 0, "Spec X", 0.7 * rho)
    ret = m.RespCombo.SetCaseList(comboName, 0, "Spec Y", -1 * 0.7 *
0.3 * rho)

    comboName = "ASD N24"
    ret = m.RespCombo.Add(comboName, comboType)
    ret = m.RespCombo.SetCaseList(comboName, 0, "Dead", 0.6 - 0.7 *
0.2 * SDS)
    ret = m.RespCombo.SetCaseList(comboName, 0, "SIDL", 0.6 - 0.7 *
0.2 * SDS)
    ret = m.RespCombo.SetCaseList(comboName, 0, "Spec X", -1 * 0.7 *
rho)
    ret = m.RespCombo.SetCaseList(comboName, 0, "Spec Y", -1 * 0.7 *
0.3 * rho)

    comboName = "ASD K1"
    ret = m.RespCombo.Add(comboName, comboType)
    ret = m.RespCombo.SetCaseList(comboName, 0, "Dead", 1 + 0.7 * 0.2
* SDS)
    ret = m.RespCombo.SetCaseList(comboName, 0, "SIDL", 1 + 0.7 * 0.2
* SDS)
    ret = m.RespCombo.SetCaseList(comboName, 0, "Spec X", 0.7 * 0.3 *
Ohm)
    ret = m.RespCombo.SetCaseList(comboName, 0, "Spec Y", 0.7 * Ohm)

    comboName = "ASD K2"
    ret = m.RespCombo.Add(comboName, comboType)
    ret = m.RespCombo.SetCaseList(comboName, 0, "Dead", 1 + 0.7 * 0.2
* SDS)
    ret = m.RespCombo.SetCaseList(comboName, 0, "SIDL", 1 + 0.7 * 0.2
* SDS)
    ret = m.RespCombo.SetCaseList(comboName, 0, "Spec X", 0.7 * 0.3 *
Ohm)
    ret = m.RespCombo.SetCaseList(comboName, 0, "Spec Y", -1 * 0.7 *
Ohm)

    comboName = "ASD K3"
    ret = m.RespCombo.Add(comboName, comboType)
```

```
    ret = m.RespCombo.SetCaseList(comboName, 0, "Dead", 1 + 0.7 * 0.2
* SDS)
    ret = m.RespCombo.SetCaseList(comboName, 0, "SIDL", 1 + 0.7 * 0.2
* SDS)
    ret = m.RespCombo.SetCaseList(comboName, 0, "Spec X", -1 * 0.7 *
0.3 * Ohm)
    ret = m.RespCombo.SetCaseList(comboName, 0, "Spec Y", 0.7 * Ohm)

    comboName = "ASD K4"
    ret = m.RespCombo.Add(comboName, comboType)
    ret = m.RespCombo.SetCaseList(comboName, 0, "Dead", 1 + 0.7 * 0.2
* SDS)
    ret = m.RespCombo.SetCaseList(comboName, 0, "SIDL", 1 + 0.7 * 0.2
* SDS)
    ret = m.RespCombo.SetCaseList(comboName, 0, "Spec X", -1 * 0.7 *
0.3 * Ohm)
    ret = m.RespCombo.SetCaseList(comboName, 0, "Spec Y", -1 * 0.7 *
Ohm)

    comboName = "ASD K5"
    ret = m.RespCombo.Add(comboName, comboType)
    ret = m.RespCombo.SetCaseList(comboName, 0, "Dead", 1 + 0.7 * 0.2
* SDS)
    ret = m.RespCombo.SetCaseList(comboName, 0, "SIDL", 1 + 0.7 * 0.2
* SDS)
    ret = m.RespCombo.SetCaseList(comboName, 0, "Spec X", 0.7 * Ohm)
    ret = m.RespCombo.SetCaseList(comboName, 0, "Spec Y", 0.7 * 0.3 *
Ohm)

    comboName = "ASD K6"
    ret = m.RespCombo.Add(comboName, comboType)
    ret = m.RespCombo.SetCaseList(comboName, 0, "Dead", 1 + 0.7 * 0.2
* SDS)
    ret = m.RespCombo.SetCaseList(comboName, 0, "SIDL", 1 + 0.7 * 0.2
* SDS)
    ret = m.RespCombo.SetCaseList(comboName, 0, "Spec X", -1 * 0.7 *
Ohm)
    ret = m.RespCombo.SetCaseList(comboName, 0, "Spec Y", 0.7 * 0.3 *
Ohm)

    comboName = "ASD K7"
    ret = m.RespCombo.Add(comboName, comboType)
    ret = m.RespCombo.SetCaseList(comboName, 0, "Dead", 1 + 0.7 * 0.2
* SDS)
    ret = m.RespCombo.SetCaseList(comboName, 0, "SIDL", 1 + 0.7 * 0.2
* SDS)
    ret = m.RespCombo.SetCaseList(comboName, 0, "Spec X", 0.7 * Ohm)
    ret = m.RespCombo.SetCaseList(comboName, 0, "Spec Y", -1 * 0.7 *
0.3 * Ohm)

    comboName = "ASD K8"
    ret = m.RespCombo.Add(comboName, comboType)
    ret = m.RespCombo.SetCaseList(comboName, 0, "Dead", 1 + 0.7 * 0.2
* SDS)
    ret = m.RespCombo.SetCaseList(comboName, 0, "SIDL", 1 + 0.7 * 0.2
* SDS)
    ret = m.RespCombo.SetCaseList(comboName, 0, "Spec X", -1 * 0.7 *
Ohm)
```

```
765         ret = m.RespCombo.SetCaseList(comboName, 0, "Spec Y", -1 * 0.7 *
766     0.3 * Ohm)
767
768         comboName = "ASD K9"
769         ret = m.RespCombo.Add(comboName, comboType)
770         ret = m.RespCombo.SetCaseList(comboName, 0, "Dead", 1 + 0.525 *
771     0.2 * SDS)
772         ret = m.RespCombo.SetCaseList(comboName, 0, "SIDL", 1 + 0.525 *
773     0.2 * SDS)
774         ret = m.RespCombo.SetCaseList(comboName, 0, "Live", 0.75)
775         ret = m.RespCombo.SetCaseList(comboName, 0, "Spec X", 0.525 * 0.3
776     * Ohm)
777         ret = m.RespCombo.SetCaseList(comboName, 0, "Spec Y", 0.525 * Ohm)
778
779         comboName = "ASD K10"
780         ret = m.RespCombo.Add(comboName, comboType)
781         ret = m.RespCombo.SetCaseList(comboName, 0, "Dead", 1 + 0.525 *
782     0.2 * SDS)
783         ret = m.RespCombo.SetCaseList(comboName, 0, "SIDL", 1 + 0.525 *
784     0.2 * SDS)
785         ret = m.RespCombo.SetCaseList(comboName, 0, "Live", 0.75)
786         ret = m.RespCombo.SetCaseList(comboName, 0, "Spec X", 0.525 * 0.3
787     * Ohm)
788         ret = m.RespCombo.SetCaseList(comboName, 0, "Spec Y", -1 * 0.525 *
789     Ohm)
790
791         comboName = "ASD K11"
792         ret = m.RespCombo.Add(comboName, comboType)
793         ret = m.RespCombo.SetCaseList(comboName, 0, "Dead", 1 + 0.525 *
794     0.2 * SDS)
795         ret = m.RespCombo.SetCaseList(comboName, 0, "SIDL", 1 + 0.525 *
796     0.2 * SDS)
797         ret = m.RespCombo.SetCaseList(comboName, 0, "Live", 0.75)
798         ret = m.RespCombo.SetCaseList(comboName, 0, "Spec X", -1 * 0.525 *
799     0.3 * Ohm)
800         ret = m.RespCombo.SetCaseList(comboName, 0, "Spec Y", 0.525 * Ohm)
801
802         comboName = "ASD K12"
803         ret = m.RespCombo.Add(comboName, comboType)
804         ret = m.RespCombo.SetCaseList(comboName, 0, "Dead", 1 + 0.525 *
805     0.2 * SDS)
806         ret = m.RespCombo.SetCaseList(comboName, 0, "SIDL", 1 + 0.525 *
807     0.2 * SDS)
808         ret = m.RespCombo.SetCaseList(comboName, 0, "Live", 0.75)
809         ret = m.RespCombo.SetCaseList(comboName, 0, "Spec X", -1 * 0.525 *
810     0.3 * Ohm)
811         ret = m.RespCombo.SetCaseList(comboName, 0, "Spec Y", -1 * 0.525 *
812     Ohm)
813
814         comboName = "ASD K13"
815         ret = m.RespCombo.Add(comboName, comboType)
816         ret = m.RespCombo.SetCaseList(comboName, 0, "Dead", 1 + 0.525 *
817     0.2 * SDS)
818         ret = m.RespCombo.SetCaseList(comboName, 0, "SIDL", 1 + 0.525 *
819     0.2 * SDS)
820         ret = m.RespCombo.SetCaseList(comboName, 0, "Live", 0.75)
821         ret = m.RespCombo.SetCaseList(comboName, 0, "Spec X", 0.525 * Ohm)
```

```
822         ret = m.RespCombo.SetCaseList(comboName, 0, "Spec Y", 0.525 * 0.3
823     * Ohm)
824
825         comboName = "ASD K14"
826         ret = m.RespCombo.Add(comboName, comboType)
827         ret = m.RespCombo.SetCaseList(comboName, 0, "Dead", 1 + 0.525 *
828     0.2 * SDS)
829         ret = m.RespCombo.SetCaseList(comboName, 0, "SIDL", 1 + 0.525 *
830     0.2 * SDS)
831         ret = m.RespCombo.SetCaseList(comboName, 0, "Live", 0.75)
832         ret = m.RespCombo.SetCaseList(comboName, 0, "Spec X", -1 * 0.525 *
833     Ohm)
834         ret = m.RespCombo.SetCaseList(comboName, 0, "Spec Y", 0.525 * 0.3
835     * Ohm)
836
837         comboName = "ASD K15"
838         ret = m.RespCombo.Add(comboName, comboType)
839         ret = m.RespCombo.SetCaseList(comboName, 0, "Dead", 1 + 0.525 *
840     0.2 * SDS)
841         ret = m.RespCombo.SetCaseList(comboName, 0, "SIDL", 1 + 0.525 *
842     0.2 * SDS)
843         ret = m.RespCombo.SetCaseList(comboName, 0, "Live", 0.75)
844         ret = m.RespCombo.SetCaseList(comboName, 0, "Spec X", 0.525 * Ohm)
845         ret = m.RespCombo.SetCaseList(comboName, 0, "Spec Y", -1 * 0.525 *
846     0.3 * Ohm)
847
848         comboName = "ASD K16"
849         ret = m.RespCombo.Add(comboName, comboType)
850         ret = m.RespCombo.SetCaseList(comboName, 0, "Dead", 1 + 0.525 *
851     0.2 * SDS)
852         ret = m.RespCombo.SetCaseList(comboName, 0, "SIDL", 1 + 0.525 *
853     0.2 * SDS)
854         ret = m.RespCombo.SetCaseList(comboName, 0, "Live", 0.75)
855         ret = m.RespCombo.SetCaseList(comboName, 0, "Spec X", -1 * 0.525 *
856     Ohm)
857         ret = m.RespCombo.SetCaseList(comboName, 0, "Spec Y", -1 * 0.525 *
858     0.3 * Ohm)
859
860         comboName = "ASD K17"
861         ret = m.RespCombo.Add(comboName, comboType)
862         ret = m.RespCombo.SetCaseList(comboName, 0, "Dead", 0.6 - 0.7 *
863     0.2 * SDS)
864         ret = m.RespCombo.SetCaseList(comboName, 0, "SIDL", 0.6 - 0.7 *
865     0.2 * SDS)
866         ret = m.RespCombo.SetCaseList(comboName, 0, "Spec X", 0.7 * 0.3 *
867     Ohm)
868         ret = m.RespCombo.SetCaseList(comboName, 0, "Spec Y", 0.7 * Ohm)
869
870         comboName = "ASD K18"
871         ret = m.RespCombo.Add(comboName, comboType)
872         ret = m.RespCombo.SetCaseList(comboName, 0, "Dead", 0.6 - 0.7 *
873     0.2 * SDS)
874         ret = m.RespCombo.SetCaseList(comboName, 0, "SIDL", 0.6 - 0.7 *
875     0.2 * SDS)
876         ret = m.RespCombo.SetCaseList(comboName, 0, "Spec X", 0.7 * 0.3 *
877     Ohm)
878         ret = m.RespCombo.SetCaseList(comboName, 0, "Spec Y", -1 * 0.7 *
879     Ohm)
```

```
880
881         comboName = "ASD K19"
882         ret = m.RespCombo.Add(comboName, comboType)
883         ret = m.RespCombo.SetCaseList(comboName, 0, "Dead", 0.6 - 0.7 *
884     0.2 * SDS)
885         ret = m.RespCombo.SetCaseList(comboName, 0, "SIDL", 0.6 - 0.7 *
886     0.2 * SDS)
887         ret = m.RespCombo.SetCaseList(comboName, 0, "Spec X", -1 * 0.7 *
888     0.3 * Ohm)
889         ret = m.RespCombo.SetCaseList(comboName, 0, "Spec Y", 0.7 * Ohm)
890
891         comboName = "ASD K20"
892         ret = m.RespCombo.Add(comboName, comboType)
893         ret = m.RespCombo.SetCaseList(comboName, 0, "Dead", 0.6 - 0.7 *
894     0.2 * SDS)
895         ret = m.RespCombo.SetCaseList(comboName, 0, "SIDL", 0.6 - 0.7 *
896     0.2 * SDS)
897         ret = m.RespCombo.SetCaseList(comboName, 0, "Spec X", -1 * 0.7 *
898     0.3 * Ohm)
899         ret = m.RespCombo.SetCaseList(comboName, 0, "Spec Y", -1 * 0.7 *
900     Ohm)
901
902         comboName = "ASD K21"
903         ret = m.RespCombo.Add(comboName, comboType)
904         ret = m.RespCombo.SetCaseList(comboName, 0, "Dead", 0.6 - 0.7 *
905     0.2 * SDS)
906         ret = m.RespCombo.SetCaseList(comboName, 0, "SIDL", 0.6 - 0.7 *
907     0.2 * SDS)
908         ret = m.RespCombo.SetCaseList(comboName, 0, "Spec X", 0.7 * Ohm)
909         ret = m.RespCombo.SetCaseList(comboName, 0, "Spec Y", 0.7 * 0.3 *
910     Ohm)
911
912         comboName = "ASD K22"
913         ret = m.RespCombo.Add(comboName, comboType)
914         ret = m.RespCombo.SetCaseList(comboName, 0, "Dead", 0.6 - 0.7 *
915     0.2 * SDS)
916         ret = m.RespCombo.SetCaseList(comboName, 0, "SIDL", 0.6 - 0.7 *
917     0.2 * SDS)
918         ret = m.RespCombo.SetCaseList(comboName, 0, "Spec X", -1 * 0.7 *
919     Ohm)
920         ret = m.RespCombo.SetCaseList(comboName, 0, "Spec Y", 0.7 * 0.3 *
921     Ohm)
922
923         comboName = "ASD K23"
924         ret = m.RespCombo.Add(comboName, comboType)
925         ret = m.RespCombo.SetCaseList(comboName, 0, "Dead", 0.6 - 0.7 *
926     0.2 * SDS)
927         ret = m.RespCombo.SetCaseList(comboName, 0, "SIDL", 0.6 - 0.7 *
928     0.2 * SDS)
929         ret = m.RespCombo.SetCaseList(comboName, 0, "Spec X", 0.7 * Ohm)
930         ret = m.RespCombo.SetCaseList(comboName, 0, "Spec Y", -1 * 0.7 *
931     0.3 * Ohm)
932
933         comboName = "ASD K24"
934         ret = m.RespCombo.Add(comboName, comboType)
935         ret = m.RespCombo.SetCaseList(comboName, 0, "Dead", 0.6 - 0.7 *
936     0.2 * SDS)
```

```
    ret = m.RespCombo.SetCaseList(comboName, 0, "SIDL", 0.6 - 0.7 *
0.2 * SDS)
    ret = m.RespCombo.SetCaseList(comboName, 0, "Spec X", -1 * 0.7 *
Ohm)
    ret = m.RespCombo.SetCaseList(comboName, 0, "Spec Y", -1 * 0.7 *
0.3 * Ohm)

    m.View.RefreshView 0, False

    MsgBox "Response Spectrum berhasil diterapkan", vbInformation
End Sub

```

Lampiran 4 Module 4 (JALANKAN_ETABSNYA)

```
Option Explicit

Private Function GetSapModel() As Object
    Dim h As Object, et As Object
    On Error Resume Next
    Set h = CreateObject("ETABSv1.Helper")
    If Not h Is Nothing Then Set et =
h.GetObject("CSI.ETABS.API.ETABSObject")
    On Error GoTo 0
    If Not et Is Nothing Then Set GetSapModel = et.SapModel
End Function

Public Sub JALANKAN_ETABSNYA()
    Dim m As Object, ws As Worksheet, ret As Long
    Dim fso As Object

    Set m = GetSapModel()
    If m Is Nothing Then
        MsgBox "ETABS tidak ditemukan.", vbExclamation
        Exit Sub
    End If

    Set ws = ActiveSheet
    Set fso = CreateObject("Scripting.FileSystemObject")

    Dim projectName As String, folderPath As String, filePath As
String
    projectName = Trim(ws.Range("G2").Value)
    If projectName = "" Then projectName = "Model_ETABS"

folderPath = CreateObject("WScript.Shell").SpecialFolders("Desktop") &
"\MACRO ETABS"

    If Not fso.FolderExists(folderPath) Then
        fso.CreateFolder folderPath
    End If

    filePath = folderPath & "\" & projectName & ".edb"

    Application.StatusBar = "Menyimpan file ETABS..."
    ret = m.File.Save(filePath)

    If ret <> 0 Then
        MsgBox "Gagal menyimpan file!", vbExclamation
        Application.StatusBar = False
```

```
        Exit Sub
    End If

    Application.StatusBar = "Menjalankan analisis..."
    ret = m.Analyze.RunAnalysis()

    If ret <> 0 Then
        MsgBox "Analisis gagal! Periksa model Anda.", vbExclamation
        Application.StatusBar = False
        Exit Sub
    End If

    Application.StatusBar = "Menjalankan design beton..."
    ret = m.DesignConcrete.StartDesign()

    If ret = 0 Then
        MsgBox "Analisis dan Design berhasil!" & vbCrLf & vbCrLf & _
               "File tersimpan di:" & vbCrLf & filePath, vbInformation
    Else
        MsgBox "Design gagal! Periksa hasil analisis.", vbExclamation
    End If

    Application.StatusBar = False
    Set fso = Nothing
End Sub
```

Lampiran 5 Module 5 (JOINT_REACTION)

```
Option Explicit

Public Sub JOINT_REACTION()
    On Error Resume Next
    Application.DisplayAlerts = False
    ThisWorkbook.Sheets("JOINT REACTION").Delete
    Application.DisplayAlerts = True
    On Error GoTo 0

    Dim ws As Worksheet
    Set ws = ThisWorkbook.Sheets.Add(After:=Sheets(Sheets.Count))
    ws.Name = "JOINT REACTION"

    '=== Connect to ETABS ===
    Dim etApp As ETABSv1.cOAPI
    Set etApp = GetObject(, "CSI.ETABS.API.ETABSObject")

    Dim etModel As ETABSv1.cSapModel
    Set etModel = etApp.SapModel

    '=== Header ===
    ws.Range("A1:J1").Value = Array("Label", "Joint ID", "Case/Combo",
"Type", _
                                    "FX (kN)", "FY (kN)", "FZ (kN)", _
                                    "MX (kNm)", "MY (kNm)", "MZ
(kNm)")
    With ws.Rows(1)
        .Font.Bold = True
        .Interior.Color = RGB(217, 217, 217)
        .HorizontalAlignment = xlCenter
```

```
31      End With
32
33      '=== Get all joints ===
34      Dim nJoint As Long, jointNames() As String
35      etModel.PointObj.GetNameList nJoint, jointNames
36
37      '=== Get all load cases & combos ===
38      Dim nCase As Long, caseNames() As String
39      Dim nCombo As Long, comboNames() As String
40      etModel.LoadCases.GetNameList nCase, caseNames
41      etModel.RespCombo.GetNameList nCombo, comboNames
42
43      '=== Variables ===
44      Dim i As Long, j As Long, rowNum As Long, labelCount As Long
45      Dim itemType As eItemTypeElm, nResult As Long
46      Dim obj() As String, elm() As String, lc() As String
47      Dim stType() As String, stNum() As Double
48      Dim fx() As Double, fy() As Double, fz() As Double
49      Dim mx() As Double, my() As Double, mz() As Double
50      Dim totalForce As Double
51
52      itemType = eItemTypeElm_Element
53      rowNum = 2
54
55      '=== LOOP SEMUA LOAD CASE ===
56      For i = 0 To nCase - 1
57          labelCount = 1
58          etModel.Results.Setup.DeselectAllCasesAndCombosForOutput
59          etModel.Results.Setup.SetCaseSelectedForOutput caseNames(i)
60
61          For j = 0 To nJoint - 1
62              etModel.Results.JointReact jointNames(j), itemType,
63  nResult, obj, elm, lc, stType, stNum, fx, fy, fz, mx, my, mz
64              If nResult > 0 Then
65                  totalForce = Abs(fx(0)) + Abs(fy(0)) + Abs(fz(0)) +
66  Abs(mx(0)) + Abs(my(0)) + Abs(mz(0))
67                  If totalForce > 0.000001 Then
68                      ws.Cells(rowNum, 1).Value = labelCount
69                      ws.Cells(rowNum, 2).Value = jointNames(j)
70                      ws.Cells(rowNum, 3).Value = caseNames(i)
71                      ws.Cells(rowNum, 4).Value = "Load Case"
72                      ws.Cells(rowNum, 5).Value = fx(0)
73                      ws.Cells(rowNum, 6).Value = fy(0)
74                      ws.Cells(rowNum, 7).Value = fz(0)
75                      ws.Cells(rowNum, 8).Value = mx(0)
76                      ws.Cells(rowNum, 9).Value = my(0)
77                      ws.Cells(rowNum, 10).Value = mz(0)
78                      labelCount = labelCount + 1
79                      rowNum = rowNum + 1
80                  End If
81              End If
82          Next j
83      Next i
84
85      '=== LOOP SEMUA LOAD COMBO ===
86      For i = 0 To nCombo - 1
87          labelCount = 1
88          etModel.Results.Setup.DeselectAllCasesAndCombosForOutput
```

```
        etModel.Results.Setup.SetComboSelectedForOutput comboNames(i)

        For j = 0 To nJoint - 1
            etModel.Results.JointReact jointNames(j), itemType,
nResult, obj, elm, lc, stType, stNum, fx, fy, fz, mx, my, mz
            If nResult > 0 Then
                totalForce = Abs(fx(0)) + Abs(fy(0)) + Abs(fz(0)) +
Abs(mx(0)) + Abs(my(0)) + Abs(mz(0))
                If totalForce > 0.000001 Then
                    ws.Cells(rowNum, 1).Value = labelCount
                    ws.Cells(rowNum, 2).Value = jointNames(j)
                    ws.Cells(rowNum, 3).Value = comboNames(i)
                    ws.Cells(rowNum, 4).Value = "Load Combo"
                    ws.Cells(rowNum, 5).Value = fx(0)
                    ws.Cells(rowNum, 6).Value = fy(0)
                    ws.Cells(rowNum, 7).Value = fz(0)
                    ws.Cells(rowNum, 8).Value = mx(0)
                    ws.Cells(rowNum, 9).Value = my(0)
                    ws.Cells(rowNum, 10).Value = mz(0)
                    labelCount = labelCount + 1
                    rowNum = rowNum + 1
                End If
            End If
        Next j
    Next i

    '=== Buat tabel Excel dengan filter aktif ===
    Dim lastRow As Long, tblRange As Range
    lastRow = ws.Cells(ws.Rows.Count, "A").End(xlUp).Row
    Set tblRange = ws.Range("A1:J" & lastRow)

    ws.ListObjects.Add(xlSrcRange, tblRange, , xlYes).Name =
"TblJointReactions"
    ws.ListObjects("TblJointReactions").TableStyle =
"TableStyleMedium2"

    ws.Columns.AutoFit

    MsgBox "Joint Reactions berhasil diekspor.", vbInformation
End Sub
```

Lampiran 6 Module 6 (BASE_REACTION)

```
Option Explicit

Public Sub BASE_REACTION()
    On Error Resume Next
    Application.DisplayAlerts = False
    ThisWorkbook.Sheets("BASE REACTION").Delete
    Application.DisplayAlerts = True
    On Error GoTo 0

    Dim ws As Worksheet
    Set ws = ThisWorkbook.Sheets.Add(After:=Sheets(Sheets.Count))
    ws.Name = "BASE REACTION"

    '=== Connect to ETABS ===
    Dim etApp As ETABSv1.cOAPI
```

```
16      Set etApp = GetObject(, "CSI.ETABS.API.ETABSObject")
17
18      Dim etModel As ETABSv1.cSapModel
19      Set etModel = etApp.SapModel
20
21      '=== Header ===
22      ws.Range("A1:H1").Value = Array("Label", "Case/Combo", "Type", _
23                                      "FX (kN)", "FY (kN)", "FZ (kN)", _
24                                      "MX (kNm)", "MY (kNm)")
25      With ws.Rows(1)
26          .Font.Bold = True
27          .Interior.Color = RGB(217, 217, 217)
28          .HorizontalAlignment = xlCenter
29      End With
30
31      '=== Get all load cases & combos ===
32      Dim nCase As Long, caseNames() As String
33      Dim nCombo As Long, comboNames() As String
34      etModel.LoadCases.GetNameList nCase, caseNames
35      etModel.RespCombo.GetNameList nCombo, comboNames
36
37      '=== Variables ===
38      Dim i As Long, rowNum As Long, labelCount As Long
39      Dim nResult As Long
40      Dim lc() As String, stType() As String, stNum() As Double
41      Dim fx() As Double, fy() As Double, fz() As Double
42      Dim mx() As Double, my() As Double, mz() As Double
43      Dim gx As Double, gy As Double, gz As Double
44
45      rowNum = 2
46      labelCount = 1
47
48      '=== LOOP SEMUA LOAD CASE ===
49      For i = 0 To nCase - 1
50          etModel.Results.Setup.DeselectAllCasesAndCombosForOutput
51          etModel.Results.Setup.SetCaseSelectedForOutput caseNames(i)
52
53          ' Get base reaction
54          etModel.Results.BaseReact nResult, lc, stType, stNum, fx, fy,
55  fz, mx, my, mz, gx, gy, gz
56
57          If nResult > 0 Then
58              ws.Cells(rowNum, 1).Value = labelCount
59              ws.Cells(rowNum, 2).Value = caseNames(i)
60              ws.Cells(rowNum, 3).Value = "Load Case"
61              ws.Cells(rowNum, 4).Value = fx(0)
62              ws.Cells(rowNum, 5).Value = fy(0)
63              ws.Cells(rowNum, 6).Value = fz(0)
64              ws.Cells(rowNum, 7).Value = mx(0)
65              ws.Cells(rowNum, 8).Value = my(0)
66              labelCount = labelCount + 1
67              rowNum = rowNum + 1
68          End If
69      Next i
70
71      '=== LOOP SEMUA LOAD COMBO ===
72      For i = 0 To nCombo - 1
73          etModel.Results.Setup.DeselectAllCasesAndCombosForOutput
```

```
        etModel.Results.Setup.SetComboSelectedForOutput comboNames(i)

        ' Get base reaction
        etModel.Results.BaseReact nResult, lc, stType, stNum, fx, fy,
fz, mx, my, mz, gx, gy, gz

        If nResult > 0 Then
            ws.Cells(rowNum, 1).Value = labelCount
            ws.Cells(rowNum, 2).Value = comboNames(i)
            ws.Cells(rowNum, 3).Value = "Load Combo"
            ws.Cells(rowNum, 4).Value = fx(0)
            ws.Cells(rowNum, 5).Value = fy(0)
            ws.Cells(rowNum, 6).Value = fz(0)
            ws.Cells(rowNum, 7).Value = mx(0)
            ws.Cells(rowNum, 8).Value = my(0)
            labelCount = labelCount + 1
            rowNum = rowNum + 1
        End If
    Next i

    '=== Buat tabel Excel dengan filter aktif ===
    Dim lastRow As Long, tblRange As Range
    lastRow = ws.Cells(ws.Rows.Count, "A").End(xlUp).Row

    If lastRow > 1 Then
        Set tblRange = ws.Range("A1:H" & lastRow)
        ws.ListObjects.Add(xlSrcRange, tblRange, , xlYes).Name =
"TblBaseReactions"
        ws.ListObjects("TblBaseReactions").TableStyle =
"TableStyleMedium2"
    End If

    ws.Columns.AutoFit

    MsgBox "Base Reactions berhasil diekspor.", vbInformation
End Sub
```

Lampiran 7 Module 7 (FRAME_FORCES_BEAM)

```
Option Explicit

Public Sub FRAME_FORCES_BEAM()
    On Error Resume Next
    Application.DisplayAlerts = False
    ThisWorkbook.Sheets("FRAME FORCES - BEAM").Delete
    Application.DisplayAlerts = True
    On Error GoTo 0

    Dim ws As Worksheet
    Set ws = ThisWorkbook.Sheets.Add(After:=Sheets(Sheets.Count))
    ws.Name = "FRAME FORCES - BEAM"

    '=== Connect to ETABS ===
    Dim etApp As ETABSv1.cOAPI
    Set etApp = GetObject(, "CSI.ETABS.API.ETABSObject")

    Dim etModel As ETABSv1.cSapModel
    Set etModel = etApp.SapModel

```

```
21      '=== Header ===
22      ws.Range("A1:M1").Value = Array("Label", "Beam ID", "Story",
23  "Case/Combo", "Type", _
24                                      "Location (m)", "P (kN)", "V2
25  (kN)", "V3 (kN)", _
26                                      "T (kNm)", "M2 (kNm)", "M3 (kNm)",
27  "Object")
28      With ws.Rows(1)
29          .Font.Bold = True
30          .Interior.Color = RGB(217, 217, 217)
31          .HorizontalAlignment = xlCenter
32      End With
33
34      '=== Get all frame objects ===
35      Dim nFrame As Long, frameNames() As String
36      etModel.FrameObj.GetNameList nFrame, frameNames
37
38      '=== Get all load cases & combos ===
39      Dim nCase As Long, caseNames() As String
40      Dim nCombo As Long, comboNames() As String
41      etModel.LoadCases.GetNameList nCase, caseNames
42      etModel.RespCombo.GetNameList nCombo, comboNames
43
44      '=== Variables ===
45      Dim i As Long, j As Long, k As Long, rowNum As Long, labelCount As
46  Long
47      Dim itemType As eItemTypeElm, numberResults As Long
48      Dim objectIds() As String, objectDistances() As Double
49      Dim elementIds() As String, elementDistances() As Double
50      Dim LoadCase() As String, StepType() As String, StepNum() As
51  Double
52      Dim P() As Double, V2() As Double, V3() As Double
53      Dim T() As Double, M2() As Double, M3() As Double
54
55      Dim point1 As String, point2 As String
56      Dim x1 As Double, y1 As Double, z1 As Double
57      Dim x2 As Double, y2 As Double, z2 As Double
58      Dim deltaZ As Double, deltaHor As Double
59      Dim isBeam As Boolean
60      Dim storyName As String, labelName As String
61
62      itemType = eItemTypeElm_ObjectElm  ' PENTING: Pakai ObjectElm
63  untuk Beam!
64      rowNum = 2
65
66      '=== LOOP SEMUA LOAD CASE ===
67      For i = 0 To nCase - 1
68          labelCount = 1
69          etModel.Results.Setup.DeselectAllCasesAndCombosForOutput
70          etModel.Results.Setup.SetCaseSelectedForOutput caseNames(i)
71
72          For j = 0 To nFrame - 1
73              ' Cek koordinat untuk deteksi Beam atau Column
74              On Error Resume Next
75              etModel.FrameObj.GetPoints frameNames(j), point1, point2
76              etModel.PointObj.GetCoordCartesian point1, x1, y1, z1
77              etModel.PointObj.GetCoordCartesian point2, x2, y2, z2
78
```

```
            deltaZ = Abs(z2 - z1)
            deltaHor = Sqr((x2 - x1) ^ 2 + (y2 - y1) ^ 2)

            ' Jika horizontal > vertikal = BEAM
            isBeam = (deltaHor > deltaZ)

            If isBeam Then
                ' Get story
                etModel.FrameObj.GetLabelNameAndStory frameNames(j),
labelName, storyName

                etModel.Results.FrameForce frameNames(j), itemType,
numberResults, objectIds, objectDistances, elementIds,
elementDistances, LoadCase, StepType, StepNum, P, V2, V3, T, M2, M3

                If numberResults > 0 Then
                    For k = 0 To numberResults - 1
                        ws.Cells(rowNum, 1).Value = labelCount
                        ws.Cells(rowNum, 2).Value = elementIds(k)
                        ws.Cells(rowNum, 3).Value = storyName
                        ws.Cells(rowNum, 4).Value = caseNames(i)
                        ws.Cells(rowNum, 5).Value = "Load Case"
                        ws.Cells(rowNum, 6).Value = objectDistances(k)
                        ws.Cells(rowNum, 7).Value = P(k)
                        ws.Cells(rowNum, 8).Value = V2(k)
                        ws.Cells(rowNum, 9).Value = V3(k)
                        ws.Cells(rowNum, 10).Value = T(k)
                        ws.Cells(rowNum, 11).Value = M2(k)
                        ws.Cells(rowNum, 12).Value = M3(k)
                        ws.Cells(rowNum, 13).Value = objectIds(k)
                        labelCount = labelCount + 1
                        rowNum = rowNum + 1
                    Next k
                End If
            End If
            On Error GoTo 0
        Next j
    Next i

    '=== LOOP SEMUA LOAD COMBO ===
    For i = 0 To nCombo - 1
        labelCount = 1
        etModel.Results.Setup.DeselectAllCasesAndCombosForOutput
        etModel.Results.Setup.SetComboSelectedForOutput comboNames(i)

        For j = 0 To nFrame - 1
            On Error Resume Next
            etModel.FrameObj.GetPoints frameNames(j), point1, point2
            etModel.PointObj.GetCoordCartesian point1, x1, y1, z1
            etModel.PointObj.GetCoordCartesian point2, x2, y2, z2

            deltaZ = Abs(z2 - z1)
            deltaHor = Sqr((x2 - x1) ^ 2 + (y2 - y1) ^ 2)

            isBeam = (deltaHor > deltaZ)

            If isBeam Then
```

```
                    etModel.FrameObj.GetLabelNameAndStory frameNames(j),
labelName, storyName

                    etModel.Results.FrameForce frameNames(j), itemType,
numberResults, objectIds, objectDistances, elementIds,
elementDistances, LoadCase, StepType, StepNum, P, V2, V3, T, M2, M3

                    If numberResults > 0 Then
                        For k = 0 To numberResults - 1
                            ws.Cells(rowNum, 1).Value = labelCount
                            ws.Cells(rowNum, 2).Value = elementIds(k)
                            ws.Cells(rowNum, 3).Value = storyName
                            ws.Cells(rowNum, 4).Value = comboNames(i)
                            ws.Cells(rowNum, 5).Value = "Load Combo"
                            ws.Cells(rowNum, 6).Value = objectDistances(k)
                            ws.Cells(rowNum, 7).Value = P(k)
                            ws.Cells(rowNum, 8).Value = V2(k)
                            ws.Cells(rowNum, 9).Value = V3(k)
                            ws.Cells(rowNum, 10).Value = T(k)
                            ws.Cells(rowNum, 11).Value = M2(k)
                            ws.Cells(rowNum, 12).Value = M3(k)
                            ws.Cells(rowNum, 13).Value = objectIds(k)
                            labelCount = labelCount + 1
                            rowNum = rowNum + 1
                        Next k
                    End If
                End If
                On Error GoTo 0
            Next j
        Next i

        '=== Buat tabel Excel dengan filter aktif ===
        Dim lastRow As Long, tblRange As Range
        lastRow = ws.Cells(ws.Rows.Count, "A").End(xlUp).Row

        If lastRow > 1 Then
            Set tblRange = ws.Range("A1:M" & lastRow)
            ws.ListObjects.Add(xlSrcRange, tblRange, , xlYes).Name =
"TblBeamForces"
            ws.ListObjects("TblBeamForces").TableStyle =
"TableStyleMedium2"
        End If

        ws.Columns.AutoFit

        MsgBox "Beam Forces berhasil diekspor.", vbInformation
End Sub
```

Lampiran 8 Module 8 (FRAME_FORCES_COLUMN)

```
Option Explicit

Public Sub FRAME_FORCES_COLUMN()
    On Error Resume Next
    Application.DisplayAlerts = False
    ThisWorkbook.Sheets("FRAME FORCES - COLUMN").Delete
    Application.DisplayAlerts = True
    On Error GoTo 0

```

```
10      Dim ws As Worksheet
11      Set ws = ThisWorkbook.Sheets.Add(After:=Sheets(Sheets.Count))
12      ws.Name = "FRAME FORCES - COLUMN"
13
14      '=== Connect to ETABS ===
15      Dim etApp As ETABSv1.cOAPI
16      Set etApp = GetObject(, "CSI.ETABS.API.ETABSObject")
17
18      Dim etModel As ETABSv1.cSapModel
19      Set etModel = etApp.SapModel
20
21      '=== Header ===
22      ws.Range("A1:L1").Value = Array("Label", "Column ID",
23  "Case/Combo", "Type", _
24                                      "Location (m)", "P (kN)", "V2
25  (kN)", "V3 (kN)", _
26                                      "T (kNm)", "M2 (kNm)", "M3 (kNm)",
27  "Object")
28      With ws.Rows(1)
29          .Font.Bold = True
30          .Interior.Color = RGB(217, 217, 217)
31          .HorizontalAlignment = xlCenter
32      End With
33
34      '=== Get all frame objects ===
35      Dim nFrame As Long, frameNames() As String
36      etModel.FrameObj.GetNameList nFrame, frameNames
37
38      '=== Get all load cases & combos ===
39      Dim nCase As Long, caseNames() As String
40      Dim nCombo As Long, comboNames() As String
41      etModel.LoadCases.GetNameList nCase, caseNames
42      etModel.RespCombo.GetNameList nCombo, comboNames
43
44      '=== Variables ===
45      Dim i As Long, j As Long, k As Long, rowNum As Long, labelCount As
46  Long
47      Dim itemType As eItemTypeElm, numberResults As Long
48      Dim objectIds() As String, objectDistances() As Double
49      Dim elementIds() As String, elementDistances() As Double
50      Dim LoadCase() As String, StepType() As String, StepNum() As
51  Double
52      Dim P() As Double, V2() As Double, V3() As Double
53      Dim T() As Double, M2() As Double, M3() As Double
54      Dim isColumn As Boolean
55
56      itemType = eItemTypeElm_Element
57      rowNum = 2
58
59      '=== LOOP SEMUA LOAD CASE ===
60      For i = 0 To nCase - 1
61          labelCount = 1
62          etModel.Results.Setup.DeselectAllCasesAndCombosForOutput
63          etModel.Results.Setup.SetCaseSelectedForOutput caseNames(i)
64
65          For j = 0 To nFrame - 1
66              isColumn = CheckIfColumn(etModel, frameNames(j))
67
```

```
            If isColumn Then
                etModel.Results.FrameForce frameNames(j), itemType,
numberResults, objectIds, objectDistances, elementIds,
elementDistances, LoadCase, StepType, StepNum, P, V2, V3, T, M2, M3

                If numberResults > 0 Then
                    For k = 0 To numberResults - 1
                        ws.Cells(rowNum, 1).Value = labelCount
                        ws.Cells(rowNum, 2).Value = elementIds(k)
                        ws.Cells(rowNum, 3).Value = caseNames(i)
                        ws.Cells(rowNum, 4).Value = "Load Case"
                        ws.Cells(rowNum, 5).Value =
elementDistances(k)
                        ws.Cells(rowNum, 6).Value = P(k)
                        ws.Cells(rowNum, 7).Value = V2(k)
                        ws.Cells(rowNum, 8).Value = V3(k)
                        ws.Cells(rowNum, 9).Value = T(k)
                        ws.Cells(rowNum, 10).Value = M2(k)
                        ws.Cells(rowNum, 11).Value = M3(k)
                        ws.Cells(rowNum, 12).Value = objectIds(k)
                        labelCount = labelCount + 1
                        rowNum = rowNum + 1
                    Next k
                End If
            End If
        Next j
    Next i

    '=== LOOP SEMUA LOAD COMBO ===
    For i = 0 To nCombo - 1
        labelCount = 1
        etModel.Results.Setup.DeselectAllCasesAndCombosForOutput
        etModel.Results.Setup.SetComboSelectedForOutput comboNames(i)

        For j = 0 To nFrame - 1
            isColumn = CheckIfColumn(etModel, frameNames(j))

            If isColumn Then
                etModel.Results.FrameForce frameNames(j), itemType,
numberResults, objectIds, objectDistances, elementIds,
elementDistances, LoadCase, StepType, StepNum, P, V2, V3, T, M2, M3

                If numberResults > 0 Then
                    For k = 0 To numberResults - 1
                        ws.Cells(rowNum, 1).Value = labelCount
                        ws.Cells(rowNum, 2).Value = elementIds(k)
                        ws.Cells(rowNum, 3).Value = comboNames(i)
                        ws.Cells(rowNum, 4).Value = "Load Combo"
                        ws.Cells(rowNum, 5).Value =
elementDistances(k)
                        ws.Cells(rowNum, 6).Value = P(k)
                        ws.Cells(rowNum, 7).Value = V2(k)
                        ws.Cells(rowNum, 8).Value = V3(k)
                        ws.Cells(rowNum, 9).Value = T(k)
                        ws.Cells(rowNum, 10).Value = M2(k)
                        ws.Cells(rowNum, 11).Value = M3(k)
                        ws.Cells(rowNum, 12).Value = objectIds(k)
                        labelCount = labelCount + 1
```

```
                        rowNum = rowNum + 1
                    Next k
                End If
            End If
        Next j
    Next i

    '=== Buat tabel Excel dengan filter aktif ===
    Dim lastRow As Long, tblRange As Range
    lastRow = ws.Cells(ws.Rows.Count, "A").End(xlUp).Row

    If lastRow > 1 Then
        Set tblRange = ws.Range("A1:L" & lastRow)
        ws.ListObjects.Add(xlSrcRange, tblRange, , xlYes).Name =
"TblColumnForces"
        ws.ListObjects("TblColumnForces").TableStyle =
"TableStyleMedium2"
    End If

    ws.Columns.AutoFit

    MsgBox "Column Forces berhasil diekspor.", vbInformation
End Sub

Private Function CheckIfColumn(etModel As ETABSv1.cSapModel, frameName
As String) As Boolean
    Dim point1 As String, point2 As String
    Dim x1 As Double, y1 As Double, z1 As Double
    Dim x2 As Double, y2 As Double, z2 As Double
    Dim deltaZ As Double, deltaHor As Double

    On Error Resume Next
    etModel.FrameObj.GetPoints frameName, point1, point2
    etModel.PointObj.GetCoordCartesian point1, x1, y1, z1
    etModel.PointObj.GetCoordCartesian point2, x2, y2, z2

    deltaZ = Abs(z2 - z1)
    deltaHor = Sqr((x2 - x1) ^ 2 + (y2 - y1) ^ 2)

    If deltaZ > deltaHor Then
        CheckIfColumn = True
    Else
        CheckIfColumn = False
    End If

    On Error GoTo 0
End Function
```

Lampiran 9 Module 9 (JOINT_DISPLACEMENT)

```
Option Explicit

Public Sub JOINT_DISPLACEMENT()
    On Error Resume Next
    Application.DisplayAlerts = False
    ThisWorkbook.Sheets("JOINT DISPLACEMENT").Delete
    Application.DisplayAlerts = True
    On Error GoTo 0

```

```
    Dim ws As Worksheet
    Set ws = ThisWorkbook.Sheets.Add(After:=Sheets(Sheets.Count))
    ws.Name = "JOINT DISPLACEMENT"

    '=== Connect to ETABS ===
    Dim etApp As ETABSv1.cOAPI
    Set etApp = GetObject(, "CSI.ETABS.API.ETABSObject")

    Dim etModel As ETABSv1.cSapModel
    Set etModel = etApp.SapModel

    '=== Header ===
    ws.Range("A1:J1").Value = Array("Label", "Joint ID", "Case/Combo",
"Type", _
                                    "UX (mm)", "UY (mm)", "UZ (mm)", _
                                    "RX (rad)", "RY (rad)", "RZ
(rad)")
    With ws.Rows(1)
        .Font.Bold = True
        .Interior.Color = RGB(217, 217, 217)
        .HorizontalAlignment = xlCenter
    End With

    '=== Get all joints ===
    Dim nJoint As Long, jointNames() As String
    etModel.PointObj.GetNameList nJoint, jointNames

    '=== Get all load cases & combos ===
    Dim nCase As Long, caseNames() As String
    Dim nCombo As Long, comboNames() As String
    etModel.LoadCases.GetNameList nCase, caseNames
    etModel.RespCombo.GetNameList nCombo, comboNames

    '=== Variables ===
    Dim i As Long, j As Long, rowNum As Long, labelCount As Long
    Dim itemType As eItemTypeElm, nResult As Long
    Dim obj() As String, elm() As String, lc() As String
    Dim stType() As String, stNum() As Double
    Dim ux() As Double, uy() As Double, uz() As Double
    Dim rx() As Double, ry() As Double, rz() As Double
    Dim totalDisp As Double

    itemType = eItemTypeElm_Element
    rowNum = 2

    '=== LOOP SEMUA LOAD CASE ===
    For i = 0 To nCase - 1
        labelCount = 1
        etModel.Results.Setup.DeselectAllCasesAndCombosForOutput
        etModel.Results.Setup.SetCaseSelectedForOutput caseNames(i)

        For j = 0 To nJoint - 1
            etModel.Results.JointDispl jointNames(j), itemType,
nResult, obj, elm, lc, stType, stNum, ux, uy, uz, rx, ry, rz
            If nResult > 0 Then
                totalDisp = Abs(ux(0)) + Abs(uy(0)) + Abs(uz(0)) +
Abs(rx(0)) + Abs(ry(0)) + Abs(rz(0))
                If totalDisp > 0.000001 Then
```

```
                    ws.Cells(rowNum, 1).Value = labelCount
                    ws.Cells(rowNum, 2).Value = jointNames(j)
                    ws.Cells(rowNum, 3).Value = caseNames(i)
                    ws.Cells(rowNum, 4).Value = "Load Case"
                    ws.Cells(rowNum, 5).Value = ux(0)
                    ws.Cells(rowNum, 6).Value = uy(0)
                    ws.Cells(rowNum, 7).Value = uz(0)
                    ws.Cells(rowNum, 8).Value = rx(0)
                    ws.Cells(rowNum, 9).Value = ry(0)
                    ws.Cells(rowNum, 10).Value = rz(0)
                    labelCount = labelCount + 1
                    rowNum = rowNum + 1
                End If
            End If
        Next j
    Next i

    '=== LOOP SEMUA LOAD COMBO ===
    For i = 0 To nCombo - 1
        labelCount = 1
        etModel.Results.Setup.DeselectAllCasesAndCombosForOutput
        etModel.Results.Setup.SetComboSelectedForOutput comboNames(i)

        For j = 0 To nJoint - 1
            etModel.Results.JointDispl jointNames(j), itemType,
nResult, obj, elm, lc, stType, stNum, ux, uy, uz, rx, ry, rz
            If nResult > 0 Then
                totalDisp = Abs(ux(0)) + Abs(uy(0)) + Abs(uz(0)) +
Abs(rx(0)) + Abs(ry(0)) + Abs(rz(0))
                If totalDisp > 0.000001 Then
                    ws.Cells(rowNum, 1).Value = labelCount
                    ws.Cells(rowNum, 2).Value = jointNames(j)
                    ws.Cells(rowNum, 3).Value = comboNames(i)
                    ws.Cells(rowNum, 4).Value = "Load Combo"
                    ws.Cells(rowNum, 5).Value = ux(0)
                    ws.Cells(rowNum, 6).Value = uy(0)
                    ws.Cells(rowNum, 7).Value = uz(0)
                    ws.Cells(rowNum, 8).Value = rx(0)
                    ws.Cells(rowNum, 9).Value = ry(0)
                    ws.Cells(rowNum, 10).Value = rz(0)
                    labelCount = labelCount + 1
                    rowNum = rowNum + 1
                End If
            End If
        Next j
    Next i

    '=== Buat tabel Excel dengan filter aktif ===
    Dim lastRow As Long, tblRange As Range
    lastRow = ws.Cells(ws.Rows.Count, "A").End(xlUp).Row

    If lastRow > 1 Then
        Set tblRange = ws.Range("A1:J" & lastRow)
        ws.ListObjects.Add(xlSrcRange, tblRange, , xlYes).Name =
"TblJointDisplacements"
        ws.ListObjects("TblJointDisplacements").TableStyle =
"TableStyleMedium2"
    End If
```

```
126
127         ws.Columns.AutoFit
128
129         MsgBox "Joint Displacements berhasil diekspor.", vbInformation
130     End Sub
```

Lampiran 10 UserForm 1 (FormETABSMain)

```
1     Option Explicit
2
3
4     Private Sub FrameGempa_Click()
5
6     End Sub
7
8     Private Sub Image1_BeforeDragOver(ByVal Cancel As
9     MSForms.ReturnBoolean, ByVal Data As MSForms.DataObject, ByVal X As
10    Single, ByVal Y As Single, ByVal DragState As MSForms.fmDragState,
11    ByVal Effect As MSForms.ReturnEffect, ByVal Shift As Integer)
12
13    End Sub
14
15    Private Sub UserForm_Initialize()
16        ' Set default tab
17        TabMain.Value = 0
18
19        ' Show first frame only
20        ShowFrame 0
21
22        ' Set default value untuk TextBox Jenis Tanah
23        cboJenisTanah.Value = "D"
24
25        ' Set default checkboxes di Tab RUN
26        chkCreateModel.Value = True
27        chkApplyLoads.Value = True
28        chkResponseSpectrum.Value = True
29        chkRunAnalysis.Value = True
30
31        ' Default untuk export (sesuaikan preferensi)
32        chkExportJointReaction.Value = False
33        chkExportBaseReaction.Value = False
34        chkExportBeamForces.Value = False
35        chkExportColumnForces.Value = False
36        chkExportDisplacement.Value = False
37    End Sub
38
39    ' --- TAB NAVIGATION ---
40    Private Sub TabMain_Change()
41        ShowFrame TabMain.Value
42    End Sub
43
44    Private Sub ShowFrame(tabIndex As Integer)
45        ' Hide all frames
46        FrameProyek.Visible = False
47        FrameGeometri.Visible = False
48        FrameDimensi.Visible = False
49        FrameMaterial.Visible = False
50        FrameBeban.Visible = False
```

```
    FrameGempa.Visible = False
    FrameRun.Visible = False

    ' Show selected frame
    Select Case tabIndex
        Case 0: FrameProyek.Visible = True
        Case 1: FrameGeometri.Visible = True
        Case 2: FrameDimensi.Visible = True
        Case 3: FrameMaterial.Visible = True
        Case 4: FrameBeban.Visible = True
        Case 5: FrameGempa.Visible = True
        Case 6: FrameRun.Visible = True
    End Select
End Sub
' --- AUTO CALCULATE TOTAL SIDL ---
Private Sub txtKeramik_Change()
    UpdateTotalSIDL
End Sub
Private Sub txtPlafon_Change()
    UpdateTotalSIDL
End Sub
Private Sub UpdateTotalSIDL()
    Dim keramik As Double, plafon As Double

    On Error Resume Next
    keramik = CDbl(txtKeramik.Value)
    plafon = CDbl(txtPlafon.Value)
    On Error GoTo 0

    txtTOTALSIDL.Value = Format(keramik + plafon, "0.00")
End Sub

' --- VALIDATION ---
Private Function ValidateInputs() As Boolean
    ValidateInputs = False

    ' Tab Proyek
    If Trim(txtNamaProyek.Value) = "" Then
        MsgBox "Nama proyek harus diisi!", vbExclamation, "Validasi
Input"
        TabMain.Value = 0
        txtNamaProyek.SetFocus
        Exit Function
    End If

    ' Tab Geometri
    If val(txtGridX.Value) < 2 Then
        MsgBox "Grid X harus minimal 2!", vbExclamation, "Validasi
Input"
        TabMain.Value = 1
        txtGridX.SetFocus
        Exit Function
    End If

    If val(txtGridY.Value) < 2 Then
        MsgBox "Grid Y harus minimal 2!", vbExclamation, "Validasi
Input"
        TabMain.Value = 1
```

```
109             txtGridY.SetFocus
110             Exit Function
111         End If
112
113         If val(txtJumlahLantai.Value) < 1 Then
114             MsgBox "Jumlah lantai harus minimal 1!", vbExclamation,
115     "Validasi Input"
116             TabMain.Value = 1
117             txtJumlahLantai.SetFocus
118             Exit Function
119         End If
120
121         If val(txtSpacingX.Value) <= 0 Then
122             MsgBox "Spacing X tidak valid!", vbExclamation, "Validasi
123     Input"
124             TabMain.Value = 1
125             txtSpacingX.SetFocus
126             Exit Function
127         End If
128
129         If val(txtSpacingY.Value) <= 0 Then
130             MsgBox "Spacing Y tidak valid!", vbExclamation, "Validasi
131     Input"
132             TabMain.Value = 1
133             txtSpacingY.SetFocus
134             Exit Function
135         End If
136
137         If val(txtTinggiLantai.Value) <= 0 Then
138             MsgBox "Tinggi lantai tidak valid!", vbExclamation, "Validasi
139     Input"
140             TabMain.Value = 1
141             txtTinggiLantai.SetFocus
142             Exit Function
143         End If
144
145         ' Tab Dimensi
146         If val(txtKolomLebar.Value) < 0 Then
147             MsgBox "Lebar kolom tidak valid!", vbExclamation, "Validasi
148     Input"
149             TabMain.Value = 2
150             txtKolomLebar.SetFocus
151             Exit Function
152         End If
153
154         If val(txtKolomTinggi.Value) < 0 Then
155             MsgBox "Tinggi kolom tidak valid!", vbExclamation, "Validasi
156     Input"
157             TabMain.Value = 2
158             txtKolomTinggi.SetFocus
159             Exit Function
160         End If
161
162         If val(txtBalokLebar.Value) < 0 Then
163             MsgBox "Lebar balok tidak valid!", vbExclamation, "Validasi
164     Input"
165             TabMain.Value = 2
166             txtBalokLebar.SetFocus
```

```
167             Exit Function
168         End If
169
170         If val(txtBalokTinggi.Value) < 0 Then
171             MsgBox "Tinggi balok tidak valid!", vbExclamation, "Validasi
172     Input"
173             TabMain.Value = 2
174             txtBalokTinggi.SetFocus
175             Exit Function
176         End If
177
178         If val(txtPelatTebal.Value) < 0 Then
179             MsgBox "Tebal pelat tidak valid!", vbExclamation, "Validasi
180     Input"
181             TabMain.Value = 2
182             txtPelatTebal.SetFocus
183             Exit Function
184         End If
185
186         ' Tab Material
187         If val(txtFc.Value) <= 0 Then
188             MsgBox "fc' tidak valid!", vbExclamation, "Validasi Input"
189             TabMain.Value = 3
190             txtFc.SetFocus
191             Exit Function
192         End If
193
194         If val(txtFy.Value) <= 0 Then
195             MsgBox "fy tidak valid!", vbExclamation, "Validasi Input"
196             TabMain.Value = 3
197             txtFy.SetFocus
198             Exit Function
199         End If
200
201         ' Tab Beban
202         If val(txtLiveLantai.Value) < 0 Then
203             MsgBox "Live load lantai tidak valid!", vbExclamation,
204     "Validasi Input"
205             TabMain.Value = 4
206             txtLiveLantai.SetFocus
207             Exit Function
208         End If
209
210         If val(txtLiveAtap.Value) < 0 Then
211             MsgBox "Live load atap tidak valid!", vbExclamation, "Validasi
212     Input"
213             TabMain.Value = 4
214             txtLiveAtap.SetFocus
215             Exit Function
216         End If
217
218         ' Tab Gempa
219
220         Dim jenisTanah As String
221         jenisTanah = UCase(Trim(cboJenisTanah.Value))
222
223         If jenisTanah = "" Then
```

```
224             MsgBox "Jenis tanah harus diisi!", vbExclamation, "Validasi
225   Input"
226             TabMain.Value = 5
227             cboJenisTanah.SetFocus
228             Exit Function
229         End If
230
231         ' Validasi nilai harus A, B, C, D, atau E
232         If jenisTanah <> "A" And jenisTanah <> "B" And jenisTanah <> "C"
233   And _
234            jenisTanah <> "D" And jenisTanah <> "E" Then
235             MsgBox "Jenis tanah harus A, B, C, D, atau E!" & vbCrLf & _
236                    "Anda input: " & jenisTanah, vbExclamation, "Validasi
237   Input"
238             TabMain.Value = 5
239             cboJenisTanah.SetFocus
240             Exit Function
241         End If
242
243         If val(txtPGA.Value) < 0 Then
244             MsgBox "PGA tidak valid!", vbExclamation, "Validasi Input"
245             TabMain.Value = 5
246             txtPGA.SetFocus
247             Exit Function
248         End If
249
250         If val(txtSs.Value) < 0 Then
251             MsgBox "Ss tidak valid!", vbExclamation, "Validasi Input"
252             TabMain.Value = 5
253             txtSs.SetFocus
254             Exit Function
255         End If
256
257         If val(txtS1.Value) < 0 Then
258             MsgBox "S1 tidak valid!", vbExclamation, "Validasi Input"
259             TabMain.Value = 5
260             txtS1.SetFocus
261             Exit Function
262         End If
263
264           If val(txtSds.Value) < 0 Then
265             MsgBox "Sds tidak valid!", vbExclamation, "Validasi Input"
266             TabMain.Value = 5
267             txtS1.SetFocus
268             Exit Function
269         End If
270
271           If val(txtOmega.Value) < 0 Then
272             MsgBox "Omega tidak valid!", vbExclamation, "Validasi Input"
273             TabMain.Value = 5
274             txtS1.SetFocus
275             Exit Function
276         End If
277
278         ValidateInputs = True
279   End Function
280
281   ' FILL EXCEL DARI FORM
```

```
Private Sub FillExcelFromForm()
    On Error GoTo FillError

    Dim ws As Worksheet
    Set ws = ThisWorkbook.Sheets("DATA INPUT")

    Application.ScreenUpdating = False

    ' Tab Proyek
    ws.Range("G2").Value = Trim(txtNamaProyek.Value)

    ' Tab Geometri
    ws.Range("G8").Value = CLng(txtGridX.Value)
    ws.Range("G9").Value = CLng(txtGridY.Value)
    ws.Range("G10").Value = CDbl(txtSpacingX.Value)
    ws.Range("G11").Value = CDbl(txtSpacingY.Value)
    ws.Range("G12").Value = CLng(txtJumlahLantai.Value)
    ws.Range("G13").Value = CDbl(txtTinggiLantai.Value)

    ' Tab Dimensi
    ws.Range("G20").Value = CDbl(txtKolomLebar.Value)
    ws.Range("G21").Value = CDbl(txtKolomTinggi.Value)
    ws.Range("G22").Value = CDbl(txtBalokLebar.Value)
    ws.Range("G23").Value = CDbl(txtBalokTinggi.Value)
    ws.Range("G24").Value = CDbl(txtPelatTebal.Value)

    ' Tab Material
    ws.Range("G17").Value = CDbl(txtFc.Value)
    ws.Range("G18").Value = CDbl(txtFy.Value)
    ws.Range("G25").Value = CDbl(txtDiaTulanganUtama.Value)
    ws.Range("G26").Value = CDbl(txtDiaSengkang.Value)
    ws.Range("G27").Value = CLng(txtJumlahTulangan.Value)
    ws.Range("G28").Value = CLng(txtJumlahKakiSengkang.Value)
    ws.Range("G29").Value = CDbl(txtJarakSengkang.Value)

    ' Tab Beban
    ws.Range("G33").Value = CDbl(txtLiveLantai.Value)
    ws.Range("G34").Value = CDbl(txtLiveAtap.Value)
    ws.Range("G36").Value = CDbl(txtKeramik.Value)
    ws.Range("G37").Value = CDbl(txtPlafon.Value)
    ws.Range("G38").Value = CDbl(txtTOTALSIDL.Value)
    ws.Range("G40").Value = CDbl(txtBebanDinding.Value)

    ' Tab Gempa
    ws.Range("G47").Value = UCase(Trim(cboJenisTanah.Value))
    ws.Range("G48").Value = CDbl(txtPGA.Value)
    ws.Range("G49").Value = CDbl(txtSs.Value)
    ws.Range("G50").Value = CDbl(txtS1.Value)
    ws.Range("G51").Value = CDbl(txtTL.Value)
    ws.Range("G53").Value = CDbl(txtTs.Value)
    ws.Range("G54").Value = CDbl(txtSds.Value)
    ws.Range("G57").Value = CDbl(txtOmega.Value)


    Application.ScreenUpdating = True
    Exit Sub

FillError:
```

```
    Application.ScreenUpdating = True
    MsgBox "Error saat mengisi Excel:" & vbCrLf & Err.Description,
vbCritical, "Error Fill Excel"
    Err.Raise Err.Number
End Sub
Private Sub btnRunSelected_Click()
    If Not ValidateInputs() Then Exit Sub

    If Not chkCreateModel.Value And Not chkApplyLoads.Value And _
       Not chkResponseSpectrum.Value And Not chkRunAnalysis.Value And
_
       Not chkExportJointReaction.Value And Not
chkExportBaseReaction.Value And _
       Not chkExportBeamForces.Value And Not
chkExportColumnForces.Value And _
       Not chkExportDisplacement.Value Then
        MsgBox "Tidak ada proses yang dipilih!" & vbCrLf & _
               "Silakan centang minimal 1 proses.", vbExclamation,
"Pilih Proses"
        Exit Sub
    End If

    Dim response As VbMsgBoxResult
    Dim prosesList As String
    prosesList = "Proses yang akan dijalankan:" & vbCrLf

    If chkCreateModel.Value Then prosesList = prosesList & "?Create
Model ETABS" & vbCrLf
    If chkApplyLoads.Value Then prosesList = prosesList & "? Masukkan
Pembebanan" & vbCrLf
    If chkResponseSpectrum.Value Then prosesList = prosesList & "?
Apply Response Spectrum" & vbCrLf
    If chkRunAnalysis.Value Then prosesList = prosesList & "? Run
Analysis & Design" & vbCrLf
    If chkExportJointReaction.Value Then prosesList = prosesList & "?
Export Joint Reaction" & vbCrLf
    If chkExportBaseReaction.Value Then prosesList = prosesList & "?
Export Base Reaction" & vbCrLf
    If chkExportBeamForces.Value Then prosesList = prosesList & "?
Export Beam Forces" & vbCrLf
    If chkExportColumnForces.Value Then prosesList = prosesList & "?
Export Column Forces" & vbCrLf
    If chkExportDisplacement.Value Then prosesList = prosesList & "?
Export Node Displacement" & vbCrLf

    response = MsgBox(prosesList & vbCrLf & "Lanjutkan?", _
                      vbQuestion + vbYesNo, "Konfirmasi Run Selected")

    If response = vbNo Then Exit Sub

    Call RunProcesses
End Sub

Private Sub btnRunAll_Click()
    ' Validasi
    If Not ValidateInputs() Then Exit Sub

    ' Konfirmasi
```

```
    Dim response As VbMsgBoxResult
    response = MsgBox("Apakah Anda yakin ingin menjalankan SEMUA
proses?" & vbCrLf & vbCrLf & _
                      "Proses ini akan:" & vbCrLf & _
                      "1. Mengisi data ke Excel" & vbCrLf & _
                      "2. Membuat model ETABS" & vbCrLf & _
                      "3. Masukkan pembebanan" & vbCrLf & _
                      "4. Apply response spectrum" & vbCrLf & _
                      "5. Run analisis & design" & vbCrLf & _
                      "6. Export semua hasil", _
                      vbQuestion + vbYesNo, "Konfirmasi Run All")

    If response = vbNo Then Exit Sub

    ' Set semua checkbox ke True
    chkCreateModel.Value = True
    chkApplyLoads.Value = True
    chkResponseSpectrum.Value = True
    chkRunAnalysis.Value = True
    chkExportJointReaction.Value = True
    chkExportBaseReaction.Value = True
    chkExportBeamForces.Value = True
    chkExportColumnForces.Value = True
    chkExportDisplacement.Value = True
    Call RunProcesses
End Sub

Private Sub RunProcesses()
    ' Disable buttons agar tidak bisa diklik 2x (BUKAN disable form!)
    btnRunSelected.Enabled = False
    btnRunAll.Enabled = False
    DoEvents

    On Error GoTo ErrorHandler

    Dim startTime As Double
    startTime = Timer

    ' MENGISI EXCELNYA
    Application.StatusBar = "?? Mengisi data ke Excel..."
    DoEvents
    Call FillExcelFromForm

    ' MENJALANKAN MODULES
    If chkCreateModel.Value Then
        Application.StatusBar = "??? Membuat model ETABS..."
        DoEvents
        Call EXCEL_KE_ETABS
    End If

    If chkApplyLoads.Value Then
        Application.StatusBar = "?? Memasukkan pembebanan..."
        DoEvents
        Call MASUKKAN_PEMBEBANAN
    End If

    If chkResponseSpectrum.Value Then
        Application.StatusBar = "?? Menerapkan response spectrum..."
```

```
456             DoEvents
457             Call APPLY_RESPONSE_SPECTRUM
458         End If
459
460         If chkRunAnalysis.Value Then
461             Application.StatusBar = "?? Menjalankan analisis & design..."
462             DoEvents
463             Call JALANKAN_ETABSNYA
464         End If
465
466         ' EXPORT HASIL DATA
467         If chkExportJointReaction.Value Then
468             Application.StatusBar = "?? Export joint reaction..."
469             DoEvents
470             Call JOINT_REACTION
471         End If
472
473         If chkExportBaseReaction.Value Then
474             Application.StatusBar = "?? Export base reaction..."
475             DoEvents
476             Call BASE_REACTION
477         End If
478
479         If chkExportBeamForces.Value Then
480             Application.StatusBar = "?? Export beam forces..."
481             DoEvents
482             Call FRAME_FORCES_BEAM
483         End If
484
485         If chkExportColumnForces.Value Then
486             Application.StatusBar = "?? Export column forces..."
487             DoEvents
488             Call FRAME_FORCES_COLUMN
489         End If
490
491         If chkExportDisplacement.Value Then
492             Application.StatusBar = "?? Export node displacement..."
493             DoEvents
494             Call JOINT_DISPLACEMENT
495         End If
496
497         Application.StatusBar = False
498
499         ' Re-enable buttons
500         btnRunSelected.Enabled = True
501         btnRunAll.Enabled = True
502
503         Dim elapsedTime As Double
504         elapsedTime = Timer - startTime
505
506         MsgBox "? Semua proses selesai!" & vbCrLf & vbCrLf & _
507                "Waktu proses: " & Format(elapsedTime, "0.0") & " detik" &
508     vbCrLf & vbCrLf & _
509                "Silakan cek hasil di ETABS dan sheet Excel.", _
510                vbInformation, "ETABS AUTO - Selesai"
511         Exit Sub
512
513     ErrorHandler:
```

```
    Application.StatusBar = False
    btnRunSelected.Enabled = True
    btnRunAll.Enabled = True

    MsgBox "? ERROR pada proses!" & vbCrLf & vbCrLf & _
           "Error Number: " & Err.Number & vbCrLf & _
           "Description: " & Err.Description & vbCrLf & vbCrLf & _
           "Tip: Pastikan ETABS sudah ter-install dan data input sudah
benar.", _
           vbCritical, "Error - ETABS AUTO"
End Sub

' CLOSE / TUTUP (X)
Private Sub UserForm_QueryClose(Cancel As Integer, CloseMode As
Integer)
    If CloseMode = vbFormControlMenu Then  ' User klik X
        Dim jawab As VbMsgBoxResult
        jawab = MsgBox("Apakah anda Yakin menutup Form ini??", _
                       vbQuestion + vbYesNo, "Konfirmasi Tutup")
        If jawab = vbNo Then Cancel = True
    End If
End Sub
```

Lampiran 11 ThisWorkBook (WorkBookSection)

```
Option Explicit

Private Sub Workbook_Open()
    Application.ScreenUpdating = False
    Application.WindowState = xlMinimized
    ThisWorkbook.Windows(1).Visible = True

    Load FormETABSMain
    FormETABSMain.Show vbModeless

    Application.ScreenUpdating = True
End Sub

Private Sub Workbook_Activate()
    If IsUserFormLoaded("FORMETABSMain") Then
        Application.WindowState = xlMinimized
    End If
End Sub

Private Sub Workbook_SheetActivate(ByVal Sh As Object)
    If IsUserFormLoaded("FORMETABSMain") Then
        Application.WindowState = xlMinimized
    End If
End Sub

Private Function IsUserFormLoaded(FormName As String) As Boolean
    Dim frm As Object
    IsUserFormLoaded = False
    For Each frm In VBA.UserForms
        If frm.Name = FormName Then
            IsUserFormLoaded = True
```

```
            Exit Function
        End If
    Next frm
End Function
```